Bridging the Gap: How Banks' Maturity Mismatch Shapes Monetary Policy Transmission*

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Abstract

This study examines how maturity mismatches in banks' balance sheets shape the transmission of monetary policy to credit supply. Linking supervisory data on approximately 1,800 euro area banks to loan-level credit records, we show that the role of maturity mismatches is highly 'shock-specific'. Mismatches amplify the effects of *unconventional* but not *conventional* monetary policies. Banks with larger maturity gaps reduce lending more sharply following monetary policy surprises regarding quantitative tightening (QT) because valuation losses on long-term assets negatively affect their net worth, causing tighter leverage constraints. To rationalize these findings, we develop a medium-scale New Keynesian DSGE model featuring a segmented financial sector, where intermediaries are differentiated by their maturity gaps. This framework explains the observed asymmetry: the high-mismatch banking segment is more exposed to long-duration losses that compress net worth, tighten endogenous leverage constraints, and amplify real economic effects through an investment wedge. In contrast, standard policy rate shocks, which mainly affect short-term rates, generate little heterogeneity in lending responses.

JEL codes: E32, E43, E51, E52, G21
Keywords: Monetary Policy Transmission, Maturity Mismatch, Bank Lending Channel, DSGE Model

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1 Introduction

In recent years, unconventional monetary policies have shifted from being exceptional interventions to fundamental elements of central banks' toolkits. Following an unprecedented expansion of Quantitative Easing (QE) in response to the COVID-19 pandemic in 2020, which pushed central bank balance sheets to historic highs, major economies have pivoted to a phase of Quantitative Tightening (QT) or Quantitative Normalisation (QN)[1]. This historic shift calls for a more thorough investigation of the transmission mechanisms of unconventional monetary policies, especially compared to conventional monetary policies (i.e., changes in central banks' policy rates).

This study focuses on the bank lending channel and investigates how maturity mismatches in banks' balance sheets shape the transmission of conventional and unconventional monetary policies to the credit supply. The core function of banks is maturity transformation: funding long-term assets (e.g., mortgages) with short(er)-term liabilities (e.g., deposits). This fundamentally entails a maturity mismatch in banks' balance sheets. We show that this mismatch is a key determinant of how monetary policy (especially unconventional) affects bank lending and, ultimately, real economic activity.

We assembled a new unbalanced panel of approximately 1,800 supervised euro area banks using quarterly Supervisory Reporting data. The dataset combines detailed financial statements (FINREP), regulatory ratios (COREP), and a maturity breakdown of inflows and outflows from the COREP "Maturity Ladder". This allows us to construct a bank-specific granular measure of the maturity gap. We merge these data with monthly loan-level information from the euro area credit registry (AnaCredit), linking each bank's financial data to information on its credit supply to firms. To identify exogenous monetary policy movements, we rely on high-frequency monetary policy shocks that capture both conventional and unconventional monetary policy surprises as constructed by Altavilla et al. (2019). The final dataset covers a six-and-a-half-year time span, from October 2018 to March 2025, at a monthly frequency.

Our empirical strategy applies local projections to estimate the cumulative effects of these shocks on bank lending. We interact the banks' maturity gap with each shock in our specification, while controlling for a comprehensive set of bank characteristics

[1] As referred to by the Member of the Executive Board of the ECB Isabel Schnabel in her speech at the ECB Conference on Money Markets 2025 (Schnabel (2025)).

(size, capital, liquidity, profitability, leverage, funding structure, and asset quality) and fixed effects (bank and country–sector-time fixed effects).

The results reveal a clear asymmetry: maturity mismatches amplify the transmission of unconventional monetary policy but not that of conventional monetary policy, demonstrating their highly 'shock-specific' role. Following unexpected changes in policy rates, banks with high and low maturity gaps adjust their lending similarly. In contrast, when policy shocks affect long-term rates, as in QT episodes, banks with greater maturity mismatches reduce credit supply more sharply. Quantitatively, we estimate that banks in the upper quartile of the maturity gap distribution decrease lending by an average of 0.30 percentage points more per basis point of tightening (in cumulative terms) than those in the lower quartile. This stronger contraction reflects valuation losses on long-duration assets, a tightening of leverage constraints, and reduced net worth, which jointly compress the credit supply among banks with higher maturity mismatches.

To rationalize these empirical findings, we extend the New Keynesian DSGE models of Gertler and Karadi (2011) and Gertler and Karadi (2013) by introducing segmented financial intermediaries characterized by heterogeneous maturity structures. To isolate the impact of balance sheet heterogeneity, we remain agnostic about the production side and model symmetric firms that differ only in how they are financed. We follow Allen and Gale (1994) and Gertler and Kiyotaki (2010a) by restricting specific firms to borrow exclusively from specific types of intermediaries: one segment is funded by intermediaries with a high maturity gap, and the other by intermediaries with a low maturity gap. Following a quantitative tightening shock, financial intermediaries in the high-maturity-gap segment experience severe valuation losses on their long-term assets. This decreases their net worth and disproportionately tightens their endogenous leverage constraints, creating a wider investment wedge that crowds out private lending and generates deeper, more persistent declines in investment and output. By contrast, following a standard policy rate shock, which primarily moves short-term rates, funding conditions and lending dynamics across the two financial segments remain remarkably similar, mirroring the limited heterogeneity observed in our empirical data.

We further examine the transmission of non-monetary disturbances—specifically technology and financial liquidity shocks—and find that the maturity mismatch structure creates 'systemic fragility'. The financing segment populated by high-maturity-gap financial intermediaries not only amplifies the contractionary effects of negative

shocks but also dampens the expansionary effects of positive shocks. For instance, in response to an adverse liquidity shock, the high-maturity-gap segment proves significantly more vulnerable, experiencing a sharper contraction in credit availability than its low-mismatch counterpart. Conversely, in the face of a positive technology shock, these heavily mismatched intermediaries act as a financial bottleneck; being locked into long-term assets prevents them from aggressively expanding credit, thereby stifling the potential economic boom.

Together, these findings demonstrate that the maturity structure of banks' balance sheets is an important state variable shaping the transmission of both monetary and non-monetary shocks. From a monetary policy perspective, maturity mismatch acts as a powerful amplifier of balance sheet-based unconventional policies—where transmission operates heavily through liquidity and valuation channels—but remains largely neutral under conventional, rate-based policy. Beyond monetary policy, our results underscore that financial sector heterogeneity arising from maturity mismatches is critical for macroeconomic stability. A banking system dominated by high maturity gaps inherently amplifies negative financial shocks while simultaneously constraining the credit expansion necessary to fully capitalize on positive technological developments.

By connecting detailed supervisory data with a structural model of banking behavior, this study highlights the central role of maturity transformation in monetary transmission and overall financial and economic stability. Thus, monitoring and managing maturity mismatches is essential not only for prudential oversight but also for understanding how different policy tools and macroeconomic disturbances propagate through the banking system and the broader economy.

The remainder of this paper is organized as follows. Section 2 reviews the related literature and positions this study within the broader context. Section 3 describes the supervisory and loan-level dataset used to construct the bank-specific maturity gap measure, outlines our empirical strategy based on local projections, and presents the main findings regarding the differential impacts of conventional and unconventional monetary policies. Section 4 develops the New Keynesian DSGE model featuring segmented financial intermediaries differentiated by their maturity gaps. Section 5 discusses the model's simulation results regarding the effects of monetary and non-monetary shocks. Section 6 concludes.

2 Literature Review

The recent conduct of monetary policy, which has seen significant deviations from traditional policy rules (Nakamura et al., 2025), has renewed interest in the specific transmission channels of both conventional and unconventional monetary policy tools. This context heightens the need to understand the structural features of the banking system that shape policy impact.

Banks' core function of maturity transformation—funding long-term assets with short-term, callable liabilities—is fundamentally linked to the transmission of monetary policy. The foundational theoretical work of Diamond and Dybvig (1983) established the dual nature of this activity: it is the mechanism by which banks provide liquidity, but simultaneously exposes them to fragility and runs. This inherent balance sheet structure creates a direct link to monetary policy through the interest rate risk channel (Van den Heuvel, 2002). This channel shows that when policy rates rise, banks with a large portfolio of fixed-rate long-term assets funded by short-term deposits suffer from net worth erosion (through lower net-interest margins) as funding costs rise faster than asset yields. In turn, this capital hit can force a contraction in lending, thereby amplifying the intended policy tightening.

However, this theoretical link has been met with a nuanced and seemingly contradictory empirical literature on whether maturity mismatches ultimately amplify or attenuate monetary policy. A large and growing body of work provides clear evidence of **amplification**, stemming from both sides of the balance sheet. On the asset side, Purnanandam (2007) showed that US banks with large, *unhedged* maturity gaps "cut their lending more" after rate hikes. This amplification mechanism is strongly supported by recent analyses of the 2022–2023 global tightening cycle. Using granular data, Coulier et al. (2024) find that euro area banks with a larger duration gap significantly "contract their lending relatively more when interest rates increase". This effect is economically meaningful and mitigated for banks that actively use interest rate derivatives to hedge their exposure. Separately, on the liability side, Drechsler et al. (2017) show that banks with market power in deposit markets also amplify tightening by widening deposit spreads, leading to deposit outflows and a contraction in lending.

In contrast, other studies find evidence of **attenuation**. Flannery and James (1984) provided early evidence that bank stock prices react to interest rate changes in a manner consistent with their maturity gaps. More recently, English et al. (2018) found

that banks with larger maturity gaps actually see profits rise from a steepening yield curve. Similarly, Gomez et al. (2021), using a US bank panel, found that banks with a positive "income gap" (assets repricing faster than liabilities) actually *reduced lending less* following a Fed tightening, suggesting that their balance sheet structure acted as a buffer.[2] This debate extends to unconventional policy; for example, during the negative interest rate policy (NIRP) era, high-deposit banks were unable to pass on negative rates, which squeezed their profits and perversely caused them to *reduce* lending (Heider et al., 2019).

Building on this foundation, our study resolves the apparent contradictions in the literature by making two contributions. First, we introduce a novel bank-level maturity gap indicator from the COREP Maturity Ladder to measure maturity transformation more precisely, using a wider sample of banks. Second, we demonstrate that the role of this gap in transmission is highly **shock-specific**, which explains the contradictory findings in the literature. We show that the transmission of conventional policy shocks, which mainly affect short-term rates, is rather homogeneous. Using high-frequency shocks from the EA-MPD (Altavilla et al., 2019), we find that lending responses are statistically similar across all maturity gap bins. In contrast, we find that unconventional (QE/QT) shocks, which raise long-term rates, generate substantial and persistent heterogeneity, with high-maturity-gap banks cutting credit supply significantly more than low-maturity-gap banks. Our analysis relies on a novel dataset matching euro area supervisory data (FINREP/COREP) to loan-level AnaCredit from 2018 to 2025, a period uniquely spanning the entire path from negative rates to QT. This empirical design allows us to separate level (short-end) from term-premium (long-end) news and map them into heterogeneous bank supply responses.

The related literature finds that bank equity prices react asymmetrically to interest rate shocks, consistent with our shock-specific view. English et al. (2018) show that bank equities fall with higher expected short-rate paths and steeper curves, with the balance sheet structure explaining the cross-section. Paul (2022) finds that bank equity reacts more negatively than that of non-financials to short-term rate hikes but more positively to term premia increases, especially for banks with larger maturity gaps.

[2] It is important to note that the "income gap" (a short-horizon repricing gap) used in Gomez et al. (2021) is conceptually distinct from the bank's overall duration gap. Their signs need not coincide; for example, a bank can display a positive one-year repricing gap while still holding a sizeable positive duration gap. Therefore, this finding does not inherently contradict the amplification channel discussed elsewhere.

We show that these market-price asymmetries have real consequences for the credit-quantity margin. Exploiting loan-level supply measures, we demonstrate that they translate into materially different lending paths under long-end versus short-end policy surprises.

Theoretically, recent literature has begun to connect these different findings by examining how banks actively manage maturity mismatches. While canonical DSGE models (Gertler and Kiyotaki, 2010b; Bernanke et al., 1999) often ignore this mismatch or treat it as a fixed, homogeneous parameter, newer models emphasize the dynamic role of asset duration (Wang, 2023; Varraso, 2024). For instance, Varraso (2024) shows that long periods of low interest rates push banks to "reach for yield" by buying longer-term assets, while Di Tella and Kurlat (2021) propose that banks use the maturity gap as an optimal dynamic hedging tool. Similarly, the frameworks of Gertler and Karadi (2011) and Gertler and Karadi (2013) explain how unconventional policies, such as Quantitative Easing, operate by easing constraints on banks' balance sheets. A unifying view suggests that while QE can substitute for traditional policy at the zero lower bound, reversing it (Quantitative Tightening, or QT) creates distinct challenges (Sims and Wu, 2021), highlighting the need for a theoretical framework that captures how heterogeneous maturity profiles react to different policy regimes. Relatedly, Garofalo (2024) develops a framework showing how unconventional policies rely on 'granular' intermediaries to affect the aggregate economy, a structural feature that conceptually parallels our use of segmented banking channels.

To formalize our empirical findings and bridge this empirical–theoretical gap, we develop a medium-scale New Keynesian DSGE model featuring a segmented financial sector characterized by heterogeneous maturity structures. We model symmetric firms that borrow exclusively from designated financial intermediaries: one segment funded by banks with a high maturity gap, and the other by banks with a low maturity gap. This framework delivers and explains the asymmetry we find empirically: when long rates rise via unconventional (QT) shocks, high-maturity-gap banks suffer severe valuation losses that compress their net worth, leading to tighter endogenous leverage constraints and sharper credit contractions through a wider investment wedge. Conversely, when short rates rise via conventional shocks, funding costs reprice broadly, and cross-bank heterogeneity is minimal. We further show that technology and liquidity shocks have distinct implications: high-mismatch intermediaries act as a financial bottleneck during positive technology shocks but amplify liquidity squeezes, mirroring our

reduced-form evidence. Finally, our micro-estimated gap targets discipline the model's financial block, allowing for rigorous macro-counterfactuals regarding the conventional versus unconventional monetary policy mix.

3 Empirical Analysis

For our empirical analysis, we construct a novel panel dataset covering around 1,800 supervised euro area banks and collect information from three distinct data sources: Supervisory Reporting data and euro area credit registry (AnaCredit) data from the European Central Bank (ECB), and the Euro Area Monetary Policy Event-Study Database (EA-MPD) from Altavilla et al. (2019). The final dataset is at a monthly frequency and covers a period of six and a half years, from October 2018 to March 2025. It contains the amount of loans outstanding at the bank and counterparty sector of economic activity (NACE sector) level at the end of each month from AnaCredit, a quarterly series of banks' regulatory ratios as well as balance sheet amounts, and income statement figures from Supervisory Reporting data, a measure of maturity mismatch between banks' assets and liabilities[3], and monthly monetary policy shocks constructed from the EA-MPD. In the next two subsections, we describe in more detail the maturity mismatch measure and the monetary policy shocks that we have used in our analysis, since these are crucial variables for our identification strategy. We refer the reader to Table 6 in Appendix C for more information on the construction of the other variables, which are used as controls in the context of this study.

3.1 A maturity gap measure

To test the hypothesis that banks with different levels of maturity mismatch between their assets and liabilities transmit monetary policy shocks differently, we construct a measure that proxies for the exposure of a bank's net worth to changes in interest rates.

[3]The bank balance sheet data, regulatory ratios, and the maturity gap measure are available as quarterly series. To align these with the monthly series from AnaCredit and the monetary policy shocks, we assign the quarterly data point corresponding to the end of the previous quarter to each month within that quarter. This approach ensures that the model accounts for the maturity gap and balance sheet structure of banks prior to the occurrence of the shock, thereby mitigating potential endogeneity concerns.

In finance, this measure is generally represented by the Macaulay duration (Macaulay (1938)), which applies to the context of portfolio valuation and interest rate sensitivity. Following the same logic, some studies have computed a net duration measure, known as the *duration gap*, when assessing banks' net exposure to interest rate risk (e.g., Coulier et al. (2024), Esposito et al. (2015)). Other studies have instead relied on a different indicator, the *maturity gap* Paul (2023). Although we believe that the duration gap is technically a more accurate measure of the sensitivity of banks' balance sheets to interest rate changes, we found it less suitable in the context of our analysis. This is because the duration gap is mechanically affected by changes in interest rates, since, based on its formula, risk-free rates are used to compute the maturity-weighted present value of the cash flows from assets and liabilities. Considering that our model specifications require lagged effects from monetary policy shocks, we preferred to avoid the introduction of spurious correlations in our identification strategy because of the chosen measure of banks' maturity mismatch in assets and liabilities. Therefore, we privileged the maturity gap indicator for our analysis, following Paul (2023).

We construct the maturity gap measure for our sample of banks based on the supervisory reporting data collected within the Common Reporting (COREP) framework, specifically in template C66.01 "Maturity Ladder." In this template, banks provide the amount of inflows and outflows from their assets, liabilities, and off-balance sheet items split into 21 maturity buckets (from "Overnight" to "Above 5 years"), depending on their residual maturity. We used this information to compute the maturity-weighted sum of inflows minus the maturity-weighted sum of outflows. This value is normalized by the total assets of the bank to obtain a maturity gap. For bank i in quarter t, the maturity gap is formally calculated as follows:

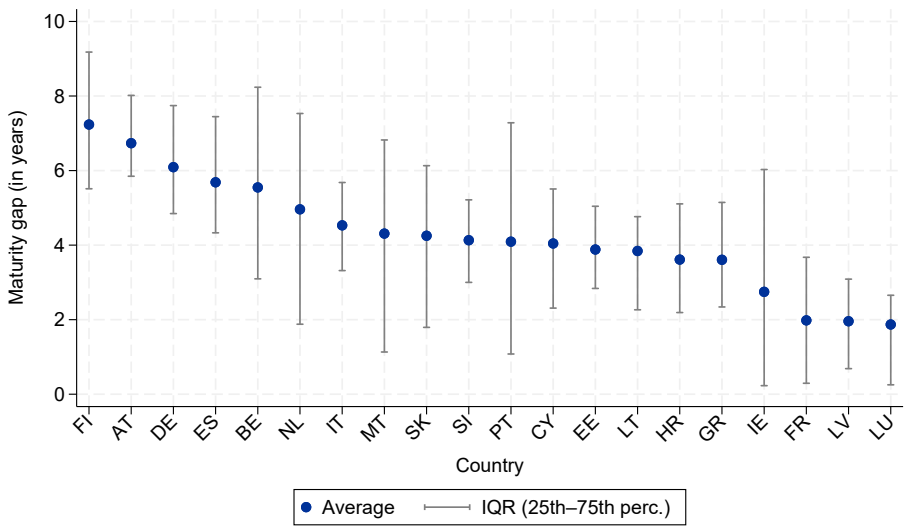
$$MatGap_{i,t} = \sum_{k=1}^{21} \frac{\tau_k(Inflows_{i,t,k} - Outflows_{i,t,k})}{TotAssets_{i,t}}$$

where τ_k is the maturity of the inflows and outflows reported in the maturity bucket k.

It is worth noting that euro area banks have different reporting requirements for the Maturity Ladder template in terms of frequency. Significant institutions are generally required to submit their reports every month, while smaller and less significant

institutions must submit them every quarter. We chose to compute quarterly series of maturity gaps to cover the largest possible sample of reporting institutions and euro area countries. Figure 1 displays the average maturity gap and the interquartile range (IQR) for all euro area countries in the sample. Although the average maturity gap for ten out of the 20 countries is within the range of 3.5-4.5 years, there is a significant variance in the maturity gap across the remaining jurisdictions: from below two years for the average bank in Luxembourg up to more than seven years for the average bank in Finland. Moreover, banks' maturity gaps vary substantially within each country. The largest IQR is recorded in Portugal (approximately 6 years) and the lowest in Austria (just above 2 years).

Figure 1: Bank maturity gap distribution within and across euro area countries



This heterogeneity likely reflects underlying differences in the composition of national banking sectors along several dimensions: size (e.g., prevalence of small cooperative banks versus large universal banks), specialization (retail-focused versus corporate or investment banking models), balance-sheet strategies, and, more generally, banks' business models. Regulatory, legal, and institutional factors could also reinforce the differences observed across countries. From the perspective of our empirical analysis (presented in Section 3.3), the presence of a broad dispersion in maturity gaps is advantageous. First, it increases the statistical power to detect the relationship between the maturity gap and our variables of interest. Second, it improves external validity: results

are less likely to be driven by a narrow subset of banks and more likely to generalize across different business models and institutional frameworks. From the perspective of our research question, the cross-country variance in the maturity gap reinforces, if anything, the interest in whether this interacts with monetary policy transmission (after controlling for bank-specific characteristics and balance sheet structure). If so, it may have policy implications. Because all euro area member states are subject to a common monetary policy, policymakers should account for banks' ex-ante maturity gaps when designing and assessing new policies. The pass-through in terms of pace and magnitude might be differentiated with potentially uneven effects across member states when it comes to credit supply, inflation, and broader macroeconomic outcomes.

3.2 The euro area monetary policy shocks

Monetary policy shocks in the euro area represent the key exogenous element in our identification strategy. To address our research question, we incorporate these shocks into our model specification by interacting them with the maturity-gap measure. Specifically, we use the Euro Area Monetary Policy Event-Study Database (EA-MPD), which provides comprehensive data on high-frequency financial market surprises in response to European Central Bank (ECB) monetary policy announcements. This database, developed by Altavilla et al. (2019), captures changes in asset prices within narrowly defined time windows around the ECB's press releases and press conferences of the ECB President. By focusing on these narrow windows, the dataset minimizes noise, thereby increasing the likelihood of capturing the causal relationships between policy announcements and observed asset price movements.

Altavilla et al. (2019) identify four monetary policy shocks, of which we primarily employ two: the *Target* shock and the *Quantitative Easing* (QE/QT) shock.[4] The target shock reflects unexpected changes at the short end of the risk-free curve, while the QT shock captures surprises affecting long-term yields and risk premia, which are typically associated with adjustments in market expectations regarding the ECB's non-standard monetary policy measures. As in the original paper, we extract these shocks by estimating a factor model through principal components applied to the matrix of

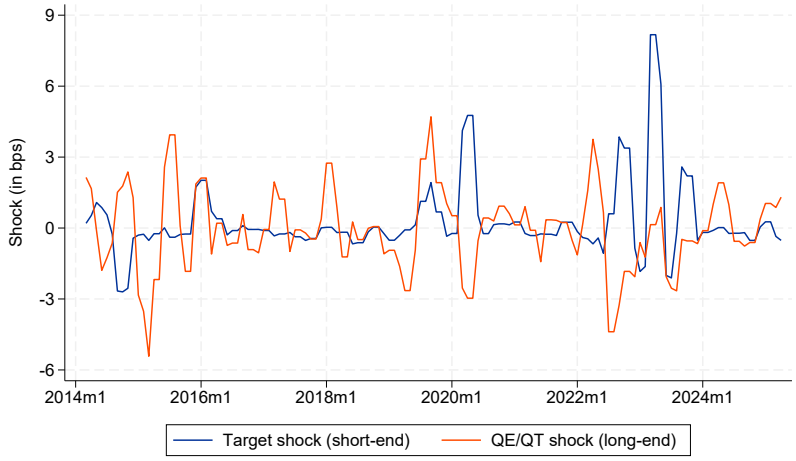
[4]In this paper, we refer to positive QE shocks as tightening surprises related to non-standard monetary policies and negative QE shocks as easing surprises. Throughout this paper, we discuss the results of positive changes in QE shocks; thus, we mostly refer to them as *QT* shocks, that is, Quantitative Tightening shocks.

yield changes and then rotating the factors to identify economically meaningful orthogonal policy shocks. This rotation is essential for disentangling the various dimensions of monetary policy surprises. The target shock is derived from the single significant factor identified in the press-release window, with its primary impact concentrated at the very short end of the yield curve (i.e., 1-month maturity) and diminishing at longer maturities.

For simplicity and consistency, we define the target shock series as the high-frequency changes in the 1-month OIS rate during the press-release window, as these changes are nearly perfectly correlated with the identified factor. Meanwhile, QT shock is associated with the third orthogonal significant factor observed in the press conference window, subject to specific restrictions. These include the shock being statistically insignificant prior to the Great Financial Crisis (before the advent of unconventional monetary policies) and the factor's loading being increasing with maturity, peaking at the long end of the yield curve (10-year maturity). The QT shock is rescaled to produce a one-unit effect on the 10-year OIS, with its sign adjusted for interpretability. The resulting monetary policy shocks, expressed in basis points, are interpreted as tightening (positive values) or easing (negative values) policy surprises. For consistency with our monthly model frequency, we extend the shocks from the ECB Governing Council meeting schedule by filling the non-meeting months with zero values.

Figure 2 illustrates the three-month moving sum of the target and QE/QT shocks since 2014. Between 2014 and 2020, QE/QT shocks were more frequent and sizeable (in absolute terms) than other types of shocks, largely reflecting the heightened focus of market participants on unconventional monetary policies as key ECB interest rates approached the effective lower bound. Conversely, from 2022 onwards, the prominence of target shocks re-emerged, driven by surprises related to the timing and pace of the ECB's tightening cycle first and easing cycle later.

Figure 2: Rotated monetary policy shocks (3-months moving sum)



In the following sections of this paper, notably in Section 4, we rely on the economic interpretation that QT shocks capture changes in market expectations concerning the ECB's unconventional monetary policies and, in particular, central bank balance sheet policies, such as the size and duration of asset purchase programs and reinvestments of the principal amounts. This is consistent with the interpretation provided by Altavilla et al. (2019) and the nature of these policies, which aim to steer rates at longer maturities. Instead, Target shocks are interpreted as surprises in market expectations concerning conventional monetary policies, that is, concerning the level of short-term yields steered by the ECB's key interest rates.

3.3 Interaction between banks' maturity gap & monetary policy transmission

Using the novel panel dataset that we constructed and enriched with monetary policy shocks and the bank-level maturity gap series, we empirically study whether banks' heterogeneity in the maturity gap matters for the transmission of monetary policy to the credit supply. The dataset we have available for this study presents the advantage of covering a time span where both a tightening and an easing cycle took place as well as multiple ECB decisions in terms of unconventional monetary policies. In addition, contrary to other studies, we have data from both significant institutions (SIs) and less significant institutions (LSIs). Thus, we can assess a broader spectrum of heterogeneous

banks and ensure that all euro-area countries are effectively represented. In this regard, the results of our study are likely to have higher external validity than analyses in which the sample was composed of relatively homogeneous banks. Moreover, as we keep the borrower's economic activity level in our data, we can also control for the loan demand component.

Table 1 summarizes the dataset used in our analysis. [5] The data are at the bank-month-borrower's economic activity level, containing an unbalanced panel of 1,803 banks and a total of 802,311 observations. The firm economic sectors considered in the sample are manufacturing, construction, retail trade, transportation, accommodation, information and communication, professional, scientific and technical activities, and administrative and support service activities. Notably, financial services and public/government-related activities were excluded. Monthly loan growth rates are constructed from outstanding loans to firms as the first difference in the logarithmic value of these amounts.

Table 1: Summary Statistics: Loan growth, monetary policy shocks and controls

	Observations	Mean	SD	P1	P25	P75	P99
Dependent variable:							
Δlog(loans)(%)	785,737	0.7	27.1	-55.0	-2.0	2.5	66.6
Monetary policy shocks:							
QE/QT shock (bps)	802,311	-0.1	0.9	-2.9	-0.4	0.2	2.5
Target shock (bps)	802,311	0.2	1.2	-1.9	-0.2	0.0	8.0
Controls:							
Maturity gap (years)	795,006	5.9	2.5	-0.5	4.6	7.7	11.0
log(total assets)	799,550	7.6	1.7	4.4	6.6	8.4	13.2
CET1 ratio (%)	800,993	17.7	6.7	10.3	14.2	19.1	47.1
Liq. cov. ratio (%)	800,546	236.2	232.4	74.3	141.8	228.0	1626.2
Leverage ratio (%)	800,870	9.4	3.7	3.7	7.4	10.6	24.4
Return on assets (%)	797,268	0.4	0.6	-1.9	0.1	0.6	2.3
NPL ratio (%)	791,271	3.4	5.1	0.1	1.3	3.4	36.6
Loan-to-deposit ratio (%)	792,899	162.4	1063.5	24.5	72.5	104.4	1893.3
Deposit ratio (%)	792,915	68.4	17.6	2.0	64.7	78.9	88.8

To further understand the structural differences in bank balance sheets, we investi-

[5] Minimum and maximum values are excluded from the summary statistics because in our analysis we winsorised the data to attenuate potential outlier effects. Therefore, the 1st and 99th percentiles can be respectively considered as the minimum and maximum values of our dataset.

gate the heterogeneity between banks with different levels of maturity transformation. Table 2 provides summary statistics split between banks with a high maturity gap and those with a low maturity gap.

There are notable differences in funding structures and liquidity positions between the two groups. High maturity gap banks appear to rely more heavily on traditional stable funding, exhibiting a significantly higher mean deposit ratio (74.1%) compared to low maturity gap banks (55.1%). Conversely, low maturity gap banks display a much higher average loan-to-deposit ratio (343.0%) compared to the high gap group (96.5%), suggesting a greater reliance on non-deposit funding sources to support their lending activities. Furthermore, low maturity gap banks maintain substantially larger liquidity buffers, with a mean liquidity coverage ratio (LCR) of 284.4%, whereas high maturity gap banks hold an average LCR of 195.3%.

Table 2: Summary Statistics: Low vs high maturity gap banks

	Observations	Mean	SD	P25	P75
High maturity gap banks:					
Δ log(loans)(%)	195,368	0.7	23.3	-2.0	2.4
Maturity gap (years)	198,762	8.7	0.9	8.0	9.1
log(total assets)	198,762	7.4	1.4	6.5	8.3
CET1 ratio (%)	198,738	17.5	5.0	14.5	18.7
Liq. cov. ratio (%)	198,762	195.3	145.2	136.9	199.0
Leverage ratio (%)	198,639	9.6	3.1	7.9	10.7
Return on assets (%)	198,576	0.4	0.6	0.1	0.7
NPL ratio (%)	198,456	2.2	2.2	1.1	2.7
Loan-to-deposit ratio (%)	198,395	96.5	305.9	78.3	103.2
Deposit ratio (%)	198,395	74.1	11.8	70.0	80.0
Low maturity gap banks:					
Δ log(loans)(%)	193,185	0.6	35.1	-2.1	2.5
Maturity gap (years)	198,770	2.4	1.6	1.1	3.8
log(total assets)	198,770	8.1	1.9	6.9	8.9
CET1 ratio (%)	198,724	18.6	9.5	13.8	19.5
Liq. cov. ratio (%)	198,350	284.4	288.4	149.6	282.2
Leverage ratio (%)	198,703	9.3	4.9	6.4	10.4
Return on assets (%)	196,985	0.4	0.8	0.1	0.7
NPL ratio (%)	191,361	5.3	8.2	1.4	4.7
Loan-to-deposit ratio (%)	195,491	343.0	1953.3	69.4	123.9
Deposit ratio (%)	195,507	55.1	24.6	40.0	74.5

We run a preliminary analysis by regressing loan growth rates over a selected set of fixed effects. The aim of this exercise is to understand whether banks' heterogeneity

is a relevant dimension to explain the observed loan growth rates. For instance, if most of the variation in loan growth rates is found to be explained by the borrower's (firm) sector-month fixed effects, and only a small residual part by banks' fixed effects (with or without time interactions), we could argue that banks' heterogeneity is a less interesting dimension to look at and that loan demand is a major driver of loan growth. However, the results in Table 3 suggest the opposite. The regression that includes only firm sector-time fixed effects (column 6) covers a negligible share of the loan growth variance ($R^2 = 0.43\%$), while bank-time fixed effects explain a quarter of the entire variance ($R^2 = 24.99\%$, column 5). The interactions between cross-sectional dimensions and time seem necessary to explain lending dynamics. This can be inferred from the result in column 4: including bank fixed effects, time fixed effects, and borrower's economic activity fixed effects without interactions yields a small R^2 of 0.43%. It could be argued that the outcome, including bank-time fixed effects, embeds country-specific dynamics rather than bank idiosyncratic ones. To test this hypothesis, we consider two additional specifications in columns 8 and 9, including country (i.e., bank location) $\times$ time fixed effects and country-time-borrower's sector fixed effects. The results are an R^2 of 0.77% for the former specification and an R^2 of 2.78% for the latter (compared to an ($R^2 = 24.99\%$ for the bank-time fixed effects regression). Therefore, while country-specific dynamics have non-negligible effects on loan growth rates, these outcomes suggest that they are largely outweighed by bank idiosyncratic reactions to shocks. We also check whether the relationship between banks and specific firm sectors explains a relevant share of the variation in loan growth rates (see column 10). For instance, firm sectors with consistently higher loan demand may be tied to a specific subset of banks, explaining the differences in loan growth observed across banks, time, and firm sectors. While the bank-firm sector dimension is non-negligible (R^2 of 1.43%), the bank-firm sector ties do not explain a large part of the story.

We can conclude from this preliminary analysis that the bank-time dimension is the most interesting to explore when explaining the dynamics of loan growth rates. Specifically, the idiosyncratic features of banks seem to matter in explaining their credit supply following shocks that hit them.

Table 3: R^2 of loan growth regressions on selected fixed effects

R^2 (in %)	0.29	0.04	0.10	0.43	24.99	0.43	25.32	0.77	2.78	1.43
Bank FE	x			x						
Firm sector FE		x		x						
Time FE			x	x						x
Bank × Time FE					x		x			
Firm sector × Time FE						x	x			
Country × Time FE								x		
Country × Time × Firm sector FE									x	
Bank × Firm sector FE										x

Following the above result, we focus on assessing the relevance of banks' maturity mismatch, among other banks' characteristics, in the transmission of monetary policy shocks to lending rates. We use an econometric specification that relies on the maturity gap measure as a proxy for banks' maturity mismatch in assets and liabilities and on the exogenous target and QT shocks described in the previous subsection. Specifically, we identify the effect of these monetary policy shocks on loan growth based on local projections, as in Jordà (2005). We regress the cumulative loan growth rate on each monetary policy shock interacted with the banks' maturity gap in the quarter before the shock materialized and a set of controls for bank size, liquidity, profitability, capitalization, leverage, asset quality, and funding structure. We include country-month-firm sector fixed effects and bank fixed effects. With the former fixed effects, we control for country-specific developments that, on average, affect all banks within a country in a similar way — such as regulatory, legal, or institutional changes, country-level shifts in funding conditions, or changes in competitive pressure — and for sector-specific fluctuations in loan demand within the bank's country, including industry-level shocks that influence firms' borrowing volumes. With bank fixed effects, we instead control for any time-invariant bank characteristics that explain loan growth levels and are potentially correlated with the maturity gap. We run the local projections on a two-year horizon (i.e., 24 months). We allow for the lagged effects of monetary policy shocks on cumulative loan growth by including all interactions with lagged shocks of up to 12 months. We use standard errors clustered at the bank and borrower sector-time levels. In summary, the regression specification is as follows:

$$\begin{aligned}\Delta y_{i,s,t+h} = \alpha_i + \lambda_{c,s,t} + \gamma_h GAP_{i,t-1} + \theta_{1h} X_{i,t-1} \\ + \sum_{l=0}^{12} \delta_h^{(l)} (MP_{t-l} \times GAP_{i,t-l-1}) + \sum_{l=0}^{12} \theta_{2h}^{(l)} (MP_{t-l} \times X_{i,t-l-1}) + \varepsilon_{i,s,t+h}\end{aligned} \tag{1}$$

where,

- $\Delta y_{i,s,t+h}$ is the cumulative bank-firm sector level loan growth rate between t and $t+h$, for $h = (0, ..., 24)$
- MP_{t-l} is the considered monetary policy shock (either target or QT) at lag $l = (0, ..., 12)$
- $GAP_{i,t-l-1}$ is the lagged maturity gap, at lag $l = (0, ..., 12)$
- $X_{i,t-1}$ is the vector of lagged bank-level controls
- α_i are the bank fixed effects, $\lambda_{c,s,t}$ are the country-sector-time fixed effects

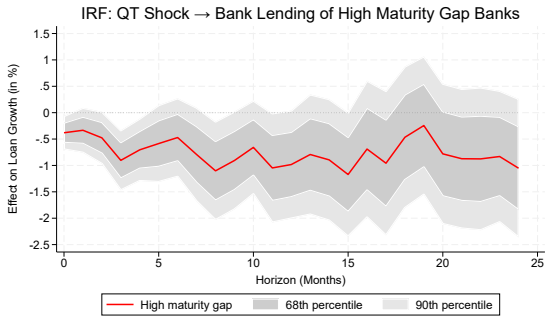
Since local projections require the h-period lead value of the loans outstanding to be in the dataset, as h increases, the number of observations in each sub-regression decreases. This can introduce a compositional effect that may produce instability in our estimates over longer horizons. We perform our baseline regressions using the full sample of observations, acknowledging this potential issue, which may produce noisier impulse responses. To ensure that our results are not driven by this compositional effect, we conduct a robustness check using a restricted sample that includes only observations for which we have a lead of loans outstanding up to the horizon $h = 24$.[6] From the specification above, we are interested in the estimated values for $\delta_h^{(0)}$. These represent the nonlinear impact of the monetary policy shock on the cumulative loan growth up to the horizon h attributable to banks' heterogeneity in their maturity gap. In terms of interpretation, a negative estimate $\hat{\delta}_h^{(0)}$ would suggest that, under a monetary policy tightening, banks with higher maturity gaps would contract their lending more than banks with lower maturity gaps and, under a monetary policy easing, banks with higher maturity gaps would expand their lending more than banks with lower maturity gaps. A positive estimate would imply the opposite interpretation.

In our baseline results using the full sample, we find a negative $\hat{\delta}_h^{(0)}$ associated with an unconventional monetary policy shock (a QT shock), as illustrated in figures 3a, 3b and 3c. We show the implied difference in cumulative loan growth for banks at the 25th versus 75th percentile of the maturity gap distribution in our sample (between three and four years of difference in the maturity gap), following a one-basis point tightening

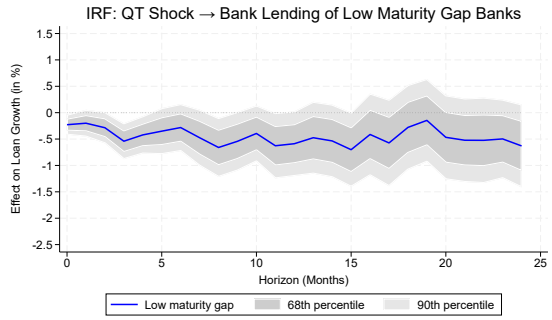
[6]The restricted sample contains 471,215 observations and ensures a constant estimation sample across all horizons. The results for this restricted sample are provided in Appendix C.

shock from unconventional policies (QT shock). Across the 24-month horizon, the difference in the cumulative impact ranges between -0.10 and -0.47 percentage points – -0.30 percentage points on average over the considered horizon – and is significant at the 90% confidence level until at least a year after a QT shock (except for a few more noisy months). The corresponding results for a one-basis point tightening target shock are included in Appendix C (see figures 11a, 11b and 11c) and show that the difference in loan growth responses between banks at the 25th and 75th percentiles of the maturity gap distribution is positive and significant on impact but largely insignificant and noisy on the overall horizon, especially after 9-10 months. In Appendix C, we also report the results for the restricted sample as a robustness check. Although they display - as expected - less noisy impulse responses, the outcome is broadly similar under a QT shock (see figure 12c). Conversely, the target shock in the restricted sample yields a statistically insignificant estimate for $\hat{\delta}_h^{(0)}$ (see figure 13c). We consider the empirical results under the target shock to be less robust because they are not fully consistent between the full and restricted sample. However, they are helpful in ruling out the hypothesis that banks with high maturity gaps decrease their loan supply more strongly than banks with low maturity gaps following a conventional monetary policy tightening shock.

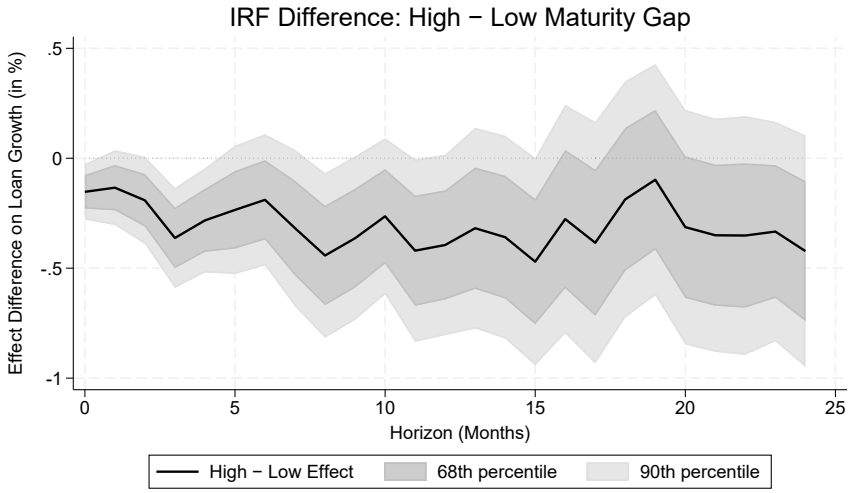
Figure 3: Analysis of bank lending responses to a QT shock, comparing high and low maturity gap banks - Full sample



(a) *Response of lending from high maturity gap banks to QT shock*



(b) *Response of lending from low maturity gap banks to QT shock*



(c) *Difference in bank lending response between banks with high vs low maturity gap under a QT shock*

Based on these findings, we can conclude that maturity gaps in banks' balance sheets *amplify* the impact of unconventional monetary policy shocks. Banks with a higher maturity gap decrease their lending more than those with a lower maturity gap after a tightening and increase their lending more compared to banks with a lower maturity gap after an easing policy. The difference is significant. In contrast, maturity gaps in banks' balance sheets do not amplify the impact of conventional monetary policy shocks, suggesting that these shocks have a rather more homogeneous effect across banks that hold different interest rate exposures.

To better understand the implications of this empirical result for real economic variables, notably inflation, economic growth, and investment, we propose a New Keynesian DSGE model with financial intermediaries. These intermediaries endogenously choose the maturity structure of their balance sheets by adjusting their average maturity through portfolio reallocation. The model and impulse response functions of the calibrated conventional and unconventional monetary policy shocks are presented in the next section.

4 The Model

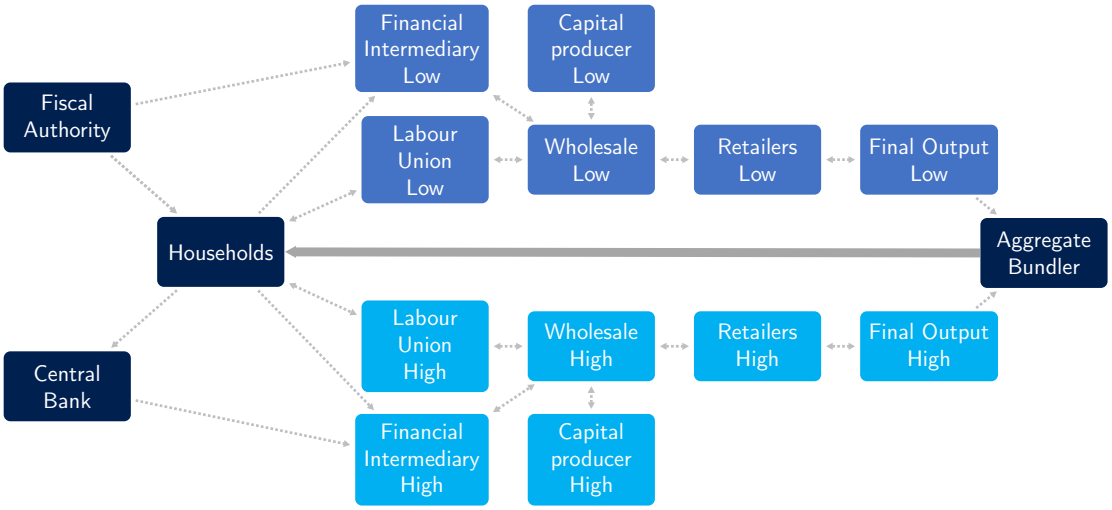


Figure 4: Stylised depiction of the model

This section outlines the core components of our model: households, labor unions, symmetric wholesale good producers, two types of segmented financial intermediaries, a fiscal authority, and a central bank. While our framework shares many similarities with canonical medium-scale DSGE models (Christiano et al., 2005; Smets and Wouters, 2007), it diverges in key ways to accommodate the segmented maturity structure of the financial sector. Below, we summarize the main features and innovative components of the framework.

Wholesale good producers (firms) use perpetual bonds with geometrically decaying coupons, following Woodford (2001), to partially finance their new capital investments. Consistent with Gertler and Karadi (2011) and Gertler and Karadi (2013), financial intermediaries fund their operations using net worth and short-term debt (deposits), while holding assets consisting of long-term private loans, government bonds, and central bank reserves. Following Gertler and Karadi (2013), market segmentation prevents households from directly holding government bonds. Furthermore, a costly enforcement problem creates an endogenous leverage constraint for financial intermediaries, generating an excess return on assets. Interacting with the firms' need to finance investment through long-term bonds, this constraint generates an "investment wedge," providing the main channel through which unconventional monetary policies—Quantitative Easing and Quantitative Tightening—affect real economic activity. The central bank finances its operations by issuing interest-bearing reserves. Crucially, while financial intermediaries optimally choose their asset and liability positions subject to their constraints, their respective maturity gaps remain defined by their specific segment (low or high). A comprehensive description of the full model is provided in Appendix A.

4.1 Model Framework and Innovation

Our framework extends the New Keynesian DSGE models of Gertler and Karadi (2011, 2013); Sims and Wu (2021). Our primary innovation is to introduce a heterogeneous banking system defined by differences in maturity gaps. Instead of modeling different industrial sectors, we assume the production side is composed of symmetric firms that are identical in technology but segmented by their source of financing. Following Allen and Gale (1994) and Gertler and Kiyotaki (2010a), specific firms borrow exclusively from specific types of financial intermediaries. This creates two parallel financing channels: one served by financial intermediaries with a "High" steady-state maturity gap (36

quarters) and the other by financial intermediaries with a "Low" steady-state maturity gap (12 quarters). This setup allows us to remain agnostic about the production side itself, cleanly isolating the impact of balance sheet heterogeneity.

Production within each financing segment follows a multi-stage process. A representative wholesale goods producer (firm) combines capital and labor to produce output, which is then purchased by a continuum of retailers. These retailers repackage wholesale output and sell it to competitive final goods producers. Capital goods producers create new physical capital specific to their segment. The output from the "Low" and "High" maturity segments is then aggregated into a final consumption bundle via a Cobb-Douglas aggregator (e.g., Ghassibe 2021), which allows representative households to consume from both sources. Households supply labor to both segments, pay taxes, and save through deposits. The labor market features two layers: labor unions purchase labor from households, and a representative labor packer aggregates this differentiated labor for final production, subject to Calvo-style nominal rigidities.

The fiscal authority consumes an exogenous, stochastic amount of final output (G_t), financed by lump-sum taxes, transfers from the central bank, and issuance of nominal bonds ($B_{G,t}$). Due to the financial intermediary frictions mentioned above, Ricardian Equivalence fails, making the tax-versus-bond financing mix relevant. We allow the supply of government bonds (b_G) to fluctuate. Lump-sum taxes adjust endogenously to ensure the government's budget constraint is satisfied in each period.

The model dynamics are driven by eighteen exogenous AR(1) processes, which can be categorized into three groups: (i) Aggregate preference and policy shocks, including investment ($\epsilon_{I,t}$), consumption ($\epsilon_{c,t}$), and labor supply ($\epsilon_{l,t}$), as well as government spending (ϵ_{G_t}), government debt supply ($\epsilon_{B,t}$), central bank government bond holdings ($\epsilon_{b,t}$), and the monetary policy rate ($\epsilon_{r,t}$); (ii) Segment-specific productivity shocks ($\epsilon_{A,L,t}, \epsilon_{A,H,t}$); and (iii) Segment-specific financial shocks, comprising central bank private loan holdings ($\epsilon_{f,L,t}, \epsilon_{f,H,t}$), liquidity constraints ($\epsilon_{\theta,L,t}, \epsilon_{\theta,H,t}$), bond recoverability ($\epsilon_{D,L,t}, \epsilon_{D,H,t}$), and shocks to the maturity of private loans ($\epsilon_{\kappa^f,L,t}, \epsilon_{\kappa^f,H,t}$) and government bond maturity ($\epsilon_{\kappa^b,t}$).

4.2 The Heterogeneous Maturity Gap and Bond Structure

Before detailing the optimization problem of the financial intermediaries, we first define the structure of their balance sheets and the specific operationalization of the maturity

gap. By assigning distinct maturity profiles to the intermediaries in each segment $s \in \{H, L\}$, we can map the resulting divergences in real activity and credit supply directly back to this balance sheet heterogeneity.

4.2.1 Bond structure

We follow Woodford (2001) and model long-term bonds as perpetuities with geometrically decaying coupon payments. This structure is defined by the decay parameter, $\kappa \in [0, 1]$. We assign distinct decay parameters to private and government bonds. Private long-term bonds (loans), issued by firms, have coupons that decay at a rate of κ_t^f. A new one-unit bond issued at price Q_t obligates the firm to pay one dollar in $t+1$, κ_t^f in $t+2$, $(\kappa_t^f)^2$ in $t+3$, and so on. Government long-term bonds, issued by the fiscal authority, have coupons that decay at a rate of κ_t^b, following an identical payment structure.

This perpetual structure is analytically convenient. We only need to track the total outstanding coupon liability from the previous period (e.g., F_{t-1} for firms) and the value of new issuances ($CF_{f,t} = F_t - \kappa_t^f F_{t-1}$). The total liability evolves as $F_{t-1} = CF_{t-1} + \kappa_t^f CF_{t-2} + (\kappa_t^f)^2 CF_{t-3} + \cdots$. The total market value of all outstanding private long-term bonds is $Q_t F_{i,t}$. The same logic applies to government long-term bonds (using $B_{G,t}$, $CB_{G,t}$, κ_t^b, and $Q_{B,t}$).

4.2.2 Long-term bond maturity

The κ parameter directly determines the effective maturity or duration of each long-term bond. A κ value closer to 1 implies slower decay, and thus a longer-maturity bond. For private long-term bonds (loans), the maturity is $M_t^f = 1/(1 - \kappa_t^f)$, and for government long-term bonds, the maturity is $M_t^b = 1/(1 - \kappa_t^b)$.

4.2.3 Maturity Gap

Following the methodology of Paul (2023), we define the maturity gap as the difference between the value-weighted average maturity of an intermediary's assets and the maturity of its liabilities. This metric captures the degree of maturity transformation performed by the financial intermediary and serves as a proxy for its exposure to interest rate risk. Formally, for bank i in segment $s \in \{H, L\}$ at time t, the maturity gap is defined as:

$$\text{Maturity Gap}_{s,i,t} = \frac{M^f_{s,i,t}(Q_{s,i,t}f_{s,i,t}) + M^b_{i,t}(Q_{B,t}b_{s,i,t}) + M^{re}_{i,t}re_{s,i,t}}{Q_{s,i,t}f_{s,i,t} + Q_{B,t}b_{s,i,t} + re_{s,i,t}} - M^d_{i,t} \tag{2}$$

The first term represents the weighted average maturity of the asset side, where weights are determined by the market values of the bank's holdings in private loans ($Q_{s,i,t}f_{s,i,t}$), government bonds ($Q_{B,t}b_{s,i,t}$), and central bank reserves ($re_{s,i,t}$). Consequently, the denominator corresponds to the total market value of the bank's assets. The second term, $M^d_{i,t}$, denotes the maturity of the bank's liabilities (deposits).

In this specification, the terms $Q_{s,i,t}f_{s,i,t}$, $Q_{B,t}b_{s,i,t}$, and $re_{s,i,t}$ represent the market values of the bank's holdings in private bonds (loans), government bonds, and reserves, respectively. Consequently, the denominator constitutes the bank's total assets at time t. Regarding the maturity parameters, $M^f_{s,i,t}$ and $M^b_{i,t}$ correspond to the long-term asset maturities derived previously. For the remaining instruments, $M^{re}_{i,t}$ and $M^d_{i,t}$ denote the maturities of reserves and deposits. Consistent with the high liquidity of these instruments, we assume they are one period ($M^{re} = M^d = 1$).

Having established the mathematical structure of the assets and the maturity gap, we now turn to the financial intermediaries' optimization problem.

4.3 Financial Intermediaries

The structure of the financial intermediaries in this study follows the framework of Gertler and Karadi (2011) and Gertler and Karadi (2013). We assume that the economy is composed of two symmetric segments, indexed by $s \in \{L, H\}$. A constant number of intermediaries exists in every period for each segment. To finance their operations, these intermediaries use their net worth ($N_{s,i,t}$) and deposits collected from the households ($D_{s,i,t}$). In each period, a random portion $(1-\sigma)$ of existing intermediaries exits the market. These exiting intermediaries return accumulated net worth to their owners (households). They are immediately replaced by an equal number of new intermediaries starting with a fixed amount of funds (X_s) provided by their household owners. On the asset side of their balance sheets, intermediaries in segment s hold long-term private bonds (loans) ($F_{s,i,t}$), long-term government bonds ($B_{s,i,t}$), and interest-bearing reserves ($RE_{s,i,t}$) in the central bank. The balance sheet condition for

a representative intermediary in segment s is

$$Q_{s,t}F_{s,i,t} + Q_{B,t}B_{s,i,t} + RE_{s,i,t} = D_{s,i,t} + N_{s,i,t} \tag{3}$$

Intermediaries grow their net worth until they stochastically exit the market. For those that survive, their net worth evolves according to the following equation:

$$\begin{aligned} N_{s,i,t} =& (R^F_{s,t} - R^d_{s,t-1})Q_{s,t-1}F_{s,i,t-1} + (R^B_t - R^d_{s,t-1})Q_{B,t-1}B_{s,i,t-1} \\ &+ (R^{re}_{t-1} - R^d_{s,t-1})RE_{s,i,t-1} + R^d_{s,t-1}N_{s,i,t-1} \end{aligned} \tag{4}$$

R^{re}_{t-1} is the interest rate on reserves set by the central bank and known ahead of time. $R^d_{s,t-1}$ is the market-based interest rate for deposits in segment s. The first three terms in the equation show the excess returns earned by the financial intermediary. This is the difference between what it earns on its assets (private bonds, government bonds, and reserves) and what it must pay out in interest on the deposits it holds. The last term shows the cost savings the bank receives from using its own net worth to fund its activities, since it does not have to pay deposit interest on that money. The terms $R^F_{s,t}$ and R^B_t represent the total realized returns from holding long-term private and government bonds, respectively, and are defined as

$$R^F_{s,t} = \frac{1 + \kappa^f_{s,t}Q_{s,t}}{Q_{s,t-1}} \tag{5}$$

$$R^B_t = \frac{1 + \kappa^b_t Q_{B,t}}{Q_{B,t-1}} \tag{6}$$

An intermediary aims to maximize the expected value of its net worth until it exits, where the expected terminal net worth is discounted using the household discount factor, $\Lambda_{t,t+1}$. Let us consider an intermediary in segment s that still operates after period t. Each period it survives with probability σ and exits with probability $1-\sigma$. Therefore, the probability that it exits at $t+1$ is $1-\sigma$, at $t+2$ is $\sigma(1-\sigma)$, and so on. Accordingly, its objective is as follows:

$$V_{s,i,t} = \max(1-\sigma)E_t \sum_{j=1}^{\infty} \sigma^{j-1}\Lambda_{t,t+j}n_{s,i,t+j}, \tag{7}$$

where $n_{s,i,t} = N_{s,i,t}/P_t$ is the real net worth and P_t is the price of the final output.

Following Gertler and Karadi (2011) and Gertler and Karadi (2013), financial intermediaries face a key constraint known as the "costly enforcement problem." This problem arises because the intermediary can choose to abandon its operations and simply divert (or steal) some of the assets it manages. If this were to happen, depositors who lent money would only be able to recover a portion of their funds, whereas the intermediary would retain the rest.

Depositors must be willing to provide funds for the system to operate. This happens only if the intermediary does not have an incentive to steal assets. We refer to this specific act of diverting funds as "going into bankruptcy." Therefore:

$$V_{s,i,t} \geq \theta_{s,t}(Q_{s,t}f_{s,i,t} + \Delta_{s,t}Q_{B,t}b_{s,i,t}) \tag{8}$$

The aforementioned inequality compares intermediaries' costs and benefits. The left-hand side of the equation shows the value of operating the business honestly, while the right-hand side shows the (real) value that the intermediary can maintain if it chooses to default. If it decides to divert, the amount it gets away with depends on its assets. It can retain the stochastic fraction $\theta_{s,t}$ of its private bonds. However, for long-term government bonds, it can only retain a smaller fraction, $\theta_{s,t}\Delta_{s,t}$ (because $\Delta_{s,t}$ is 1 or less). This assumption simply implies that it is easier to divert long-term private bonds than long-term government bonds. We assume that the third asset—reserves—is fully recoverable by depositors and cannot be stolen. We treat this fraction $\theta_{s,t}$ as both stochastic and exogenous. This can be seen as a "liquidity shock" specific to segment s. When $\theta_{s,t}$ increases, the intermediary can divert a larger portion of its assets, which means that depositors would recover less. This in turn makes depositors less willing to provide funds. This reluctance to lend is what causes interest rate spreads to rise, a classic sign of a liquidity crisis. To obtain the real version of the net worth accumulation equation, we divide both sides by the aggregate price level (P_t). Note that inflation (Π_t) is defined as the current price level relative to the price level in the previous period ($\Pi_t = P_t/P_{t-1}$).

$$\begin{aligned}\Pi_t n_{s,i,t} = &\left(R^F_{s,t} - R^d_{s,t-1}\right) Q_{s,t-1}f_{s,i,t-1} + \left(R^B_t - R^d_{s,t-1}\right) Q_{B,t-1}b_{s,i,t-1} \\ &+ \left(R^{re}_{t-1} - R^d_{s,t-1}\right) re_{s,i,t-1} + R^d_{s,t-1}n_{s,i,t-1}\end{aligned} \tag{9}$$

which implies

$$\Lambda_{t,t+1}\Omega_{s,t+1}n_{s,i,t+1} = \Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}\Big[\left(R_{s,t+1}^{F} - R_{s,t}^{d}\right)Q_{s,t}f_{s,i,t} + \left(R_{t+1}^{B} - R_{s,t}^{d}\right)Q_{B,t}b_{s,i,t} + \left(R_{t}^{re} - R_{s,t}^{d}\right)re_{s,i,t} + R_{s,t}^{d}n_{s,i,t}\Big] \tag{10}$$

Therefore, an intermediary's value function can be written as

$$\begin{aligned} V_{s,i,t} &= \max(1-\sigma)E_t\Lambda_{t,t+1}n_{s,i,t+1} + \sigma E_t\Lambda_{t,t+1}V_{s,i,t+1} \\ &= \max(1-\sigma)E_t\Bigg[\Lambda_{t,t+1}\Bigg(\frac{R_{s,t+1}^{F} - R_{s,t}^{d}}{\Pi_{t+1}}Q_{s,t}f_{s,i,t} \\ &\quad + \frac{R_{t+1}^{B} - R_{s,t}^{d}}{\Pi_{t+1}}Q_{B,t}b_{s,i,t} \\ &\quad + \frac{R_{t}^{re} - R_{s,t}^{d}}{\Pi_{t+1}}re_{s,i,t} \\ &\quad + \frac{R_{s,t}^{d}}{\Pi_{t+1}}n_{s,i,t}\Bigg)\Bigg] \\ &\quad + \sigma E_t\Lambda_{t,t+1}V_{s,i,t+1} \end{aligned} \tag{11}$$

The Lagrangian incorporating the constraint is

$$\begin{aligned} \mathcal{L}_{s,i,t} = \max(1+\lambda_{s,t})E_t\Bigg[&(1-\sigma)\Lambda_{t,t+1}\Bigg(\frac{R_{s,t+1}^{F} - R_{s,t}^{d}}{\Pi_{t+1}}Q_{s,t}f_{s,i,t} + \frac{R_{t+1}^{B} - R_{s,t}^{d}}{\Pi_{t+1}}Q_{B,t}b_{s,i,t} \\ &+ \frac{R_{t}^{re} - R_{s,t}^{d}}{\Pi_{t+1}}re_{s,i,t} + \frac{R_{s,t}^{d}}{\Pi_{t+1}}n_{s,i,t}\Bigg) + \sigma\Lambda_{t,t+1}V_{s,i,t+1}\Bigg] \\ &- \lambda_{s,t}\theta_{s,t}(Q_{s,t}f_{s,i,t} + \Delta_{s,t}Q_{B,t}b_{s,i,t}) \end{aligned} \tag{12}$$

In this equation, $\lambda_{s,t}$ denotes the Lagrange multiplier for segment s. As all financial intermediaries within a segment are assumed to be identical, they behave in the same way. Consequently, they share the same set of optimality conditions, as presented below:

$$E_t\left[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}(R_{s,t+1}^{F} - R_{s,t}^{d})\right] = \frac{\lambda_{s,t}}{1+\lambda_{s,t}}\theta_{s,t} \tag{13}$$

$$E_t\left[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}(R_{t+1}^{B}-R_{s,t}^{d})\right]=\frac{\lambda_{s,t}}{1+\lambda_{s,t}}\theta_{s,t}\Delta_{s,t} \tag{14}$$

$$E_t\left[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}(R_{t}^{re}-R_{s,t}^{d})\right]=0, \tag{15}$$

where

$$\Omega_{s,t}=1-\sigma+\sigma\theta_{s,t}\phi_{s,t} \tag{16}$$

$$\phi_{s,t}=\frac{1+\lambda_{s,t}}{\theta_{s,t}}E_t[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}]R_{s,t}^{d} \tag{17}$$

Equations (13)–(15) represent the key equilibrium conditions of the model for segment s, and the term $\lambda_{s,t}\geq 0$ is the multiplier associated with the costly enforcement constraint. There are two possible outcomes based on this constraint: First, if the constraint is not active (it "does not bind"), the expected returns on all three asset types must, to a first approximation, equal the cost of funds (the deposit rate). Second, if the constraint is active (it "binds"), both long-term private and public bonds will generate excess returns over the deposit rate. The condition $\Delta_{s,t}<1$ ensures that the excess returns on long-term government bonds are lower than those on long-term private bonds. In principle, $\Delta_{s,t}$ can be treated as an external time-varying factor that separates the term premium from corporate spread. Finally, $\Omega_{s,t}$ and $\phi_{s,t}$ are auxiliary variables introduced to simplify the analysis, and we assume the intermediary's value is a linear combination of its net worth; thus, the value of an intermediary satisfies

$$V_{s,i,t}=\theta_{s,t}\phi_{s,t}n_{s,i,t} \tag{18}$$

When the constraint binds,

$$\phi_{s,t}=\frac{Q_{s,t}f_{s,i,t}+\Delta_{s,t}Q_{B,t}b_{s,i,t}}{n_{s,i,t}} \tag{19}$$

This term is called the endogenous leverage ratio. This constraint forces the financial intermediary to be levered less (using less debt) than it would ideally have chosen. Ultimately, this internal leverage constraint allows excess returns.

To find expressions for the auxiliary variables $\Omega_{s,t}$ and $\phi_{s,t}$ in Equations (16) and (17), we start by assuming that the value function is linear in net worth, as previously mentioned. We then combine this assumption with the binding constraint and the first-order conditions. Specifically, we use the proportionality of the expected excess

returns from Equations (13) and (14), which implies $E_t[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}(R_{t+1}^B - R_{s,t}^d)] = \Delta_{s,t}\, E_t[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}(R_{s,t+1}^F - R_{s,t}^d)]$, to combine asset positions $f_{s,i,t}$ and $b_{s,i,t}$. This results in the following:

$$\begin{aligned}
&E_t[\Lambda_{t,t+1}\Omega_{s,t+1}n_{s,i,t+1}] \\
=&E_t\left[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}(R_{s,t+1}^F - R_{s,t}^d)\right](Q_{s,t}f_{s,i,t} + \Delta_{s,t}Q_{B,t}b_{s,i,t}) \\
&+ E_t\left[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}\right]R_{s,t}^d n_{s,i,t} \\
&= \frac{\lambda_{s,t}}{1+\lambda_{s,t}}\theta_{s,t}\phi_{s,t}n_{s,i,t} + E_t\left[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}\right]R_{s,t}^d n_{s,i,t}
\end{aligned}$$

We rewrite the value function in Equation (11) using (16) and (18).

$$\theta_{s,t}\phi_{s,t}n_{s,i,t} = \max E_t[\Lambda_{t,t+1}n_{s,i,t+1}\Omega_{s,t+1}] = \frac{\lambda_{s,t}}{1+\lambda_{s,t}}\theta_{s,t}\phi_{s,t}n_{s,i,t} + E_t\left[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}\right]R_{s,t}^d n_{s,i,t}$$

This leads to

$$\theta_{s,t}\phi_{s,t} = \frac{\lambda_{s,t}}{1+\lambda_{s,t}}\theta_{s,t}\phi_{s,t} + E_t\left[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}\right]R_{s,t}^d$$

Simplifying,

$$\theta_{s,t}\phi_{s,t} = (1+\lambda_{s,t})E_t\left[\Lambda_{t,t+1}\Omega_{s,t+1}\Pi_{t+1}^{-1}\right]R_{s,t}^d$$

Finally, as mentioned already, the model includes two segments as opposed to one. Importantly, we assume financing segmentation (Allen and Gale, 1994, 2007; Gertler and Kiyotaki, 2010a): each intermediary can only serve its own sector. Every asset in the intermediaries' balance sheet is segment-specific; in particular, long-term private bond holdings are circumscribed to the corresponding segment.

4.4 Production

The production process involves several stages. We discuss only wholesale goods producers below, as they are central to our model; the rest of the production side is presented in Appendix A. A representative wholesale goods producer in segment s produces intermediate output ($Y_{m,s,t}$) using capital and labor. This wholesale output is sold to a continuum of retail firms in segment s, indexed by $f \in [0,1]$. These retailers simply repackage goods, such that their individual output is $Y_{s,t}(f) = Y_{m,s,t}(f)$. These

differentiated retail goods are then purchased by a competitive final goods producer that bundles them into a single final good (Y_t) using a CES aggregation function. The elasticity of substitution between retail goods is $\epsilon_p > 1$, which ultimately determines the demand curve for each retailer. A separate competitive capital goods producer also produces new physical capital (I_t).

4.4.1 Wholesale Good Producers

The representative wholesale firm in segment s produces output according to Cobb-Douglas technology:

$$Y_{m,s,t} = A_{s,t}(u_{s,t}K_{s,t-1})^{\alpha}(L_{d,s,t})^{1-\alpha} \tag{20}$$

$Y_{m,s,t}$ represents the total output produced during period t in segment s, and $L_{d,s,t}$ is the labor input used in that same period. The firm enters period t with a predetermined stock of physical capital, $K_{s,t-1}$, and chooses a utilization rate $u_{s,t}$, where higher utilization increases effective capital services but accelerates depreciation. Parameter α is a value between zero and one, which represents the capital's relative contribution to production. $A_{s,t}$ denotes overall productivity (e.g., technology) in segment s, which is exogenous and follows a stochastic process. The depreciation rate is an increasing, convex function of utilization:

$$\delta(u_{s,t}) = \delta_{0,s} + \delta_{1,s}(u_{s,t} - 1) + \frac{\delta_2}{2}(u_{s,t} - 1)^2, \tag{21}$$

where $\delta_{0,s}$ is the steady-state depreciation rate (at $u = 1$), and $\delta_{1,s}$ and δ_2 govern the sensitivity to utilization. Capital accumulates according to:

$$K_{s,t} = \hat{I}_{s,t} + \Big(1 - \delta(u_{s,t})\Big)K_{s,t-1} \tag{22}$$

We follow an approach similar to Carlstrom et al. (2017), assuming that the wholesale goods producer must issue long-term bonds (debt) to fund its purchases of new physical capital, $I_{s,t}$. However, we make one key change: unlike their model, our firm does not need to finance its entire investment. Instead, we require that the firm finance only a constant fraction ψ_s of its new investment, where ψ_s is a value between zero and one. This requirement creates a "loan-in-advance constraint," which is expressed as:

$$\psi_s P_t^k \hat{I}_{s,t} \leq Q_{s,t}CF_{m,s,t} = Q_{s,t}(F_{m,s,t} - \kappa_{s,t}^f F_{m,s,t-1}), \tag{23}$$

where P_t^k is the price at which the wholesale firm purchases the new capital.

The wholesale goods producer employs workers from a competitive spot market at nominal wage $W_{s,t}$. The firm's resulting dividend, which is also a nominal value, is

$$\begin{aligned} DIV_{m,s,t} =& P_{m,s,t} A_{s,t} (u_{s,t} K_{s,t-1})^{\alpha} L_{d,s,t}^{1-\alpha} - W_{s,t} L_{d,s,t} - P_t^k \hat{I}_{s,t} \\ & - F_{m,s,t-1} + Q_{s,t}(F_{m,s,t} - \kappa_{s,t}^f F_{m,s,t-1}) \end{aligned} \tag{24}$$

The firm's real dividend is

$$\begin{aligned} div_{m,s,t} =& p_{m,s,t} A_{s,t} (u_{s,t} K_{s,t-1})^{\alpha} L_{d,s,t}^{1-\alpha} - w_{s,t} L_{d,s,t} - p_t^k \hat{I}_{s,t} \\ & + Q_{s,t}\left(\frac{F_{m,s,t}}{P_t} - \kappa_{s,t}^f \frac{F_{m,s,t-1}}{P_{t-1}} \Pi_t^{-1}\right) - \frac{F_{m,s,t-1}}{P_{t-1}} \Pi_t^{-1} \end{aligned} \tag{25}$$

A firm's objective is to maximize the present discounted value of its real dividends. To do this, it chooses its labor ($L_{d,s,t}$), capital utilization ($u_{s,t}$), the next period's capital ($K_{s,t}$), investment ($\hat{I}_{s,t}$), and financing ($f_{m,s,t}$). Discounting is based on the households' stochastic discount factor, which is solved using a Lagrangian as follows:

$$\begin{aligned} \mathcal{L}_{m,s,t} = E_t \sum_{j=0}^{\infty} \Lambda_{t,t+j} \Bigg\{ & p_{m,s,t+j} A_{s,t+j} (u_{s,t+j} K_{s,t+j-1})^{\alpha} L_{d,s,t+j}^{1-\alpha} - w_{s,t+j} L_{d,s,t+j} - p_{t+j}^k \hat{I}_{s,t+j} \\ & + Q_{s,t+j}\left(\frac{F_{m,s,t+j}}{P_{t+j}} - \kappa_{s,t+j}^f \frac{F_{m,s,t+j-1}}{P_{t+j-1}} \Pi_{t+j}^{-1}\right) - \frac{F_{m,s,t+j-1}}{P_{t+j-1}} \Pi_{t+j}^{-1} \\ & + \nu_{1,s,t+j}\Big(\hat{I}_{s,t+j} + \big(1 - \delta(u_{s,t+j})\big) K_{s,t+j-1} - K_{s,t+j}\Big) \\ & + \nu_{2,s,t+j}\left(Q_{s,t+j}\left(\frac{F_{m,s,t+j}}{P_{t+j}} - \kappa_{s,t+j}^f \frac{F_{m,s,t+j-1}}{P_{t+j-1}} \Pi_{t+j}^{-1}\right) - \psi_s p_{t+j}^k \hat{I}_{s,t+j}\right) \Bigg\} \end{aligned} \tag{26}$$

The first-order conditions are as follows:

$$w_{s,t} = (1-\alpha) p_{m,s,t} A_{s,t} (u_{s,t} K_{s,t-1})^{\alpha} L_{d,s,t}^{-\alpha} \tag{27}$$

$$p_t^k M_{1,s,t}\Big(\delta_{1,s} + \delta_2(u_{s,t} - 1)\Big) = \alpha\, p_{m,s,t} A_{s,t} (u_{s,t} K_{s,t-1})^{\alpha-1} L_{d,s,t}^{1-\alpha} \tag{28}$$

$$p_t^k M_{1,s,t} = E_t \Lambda_{t,t+1} \Big[\alpha \, p_{m,s,t+1} A_{s,t+1} K_{s,t}^{\alpha-1} u_{s,t+1}^{\alpha} L_{d,s,t+1}^{1-\alpha} + \Big(1 - \delta(u_{s,t+1})\Big) p_{t+1}^k M_{1,s,t+1} \Big] \tag{29}$$

$$Q_{s,t} M_{2,s,t} = E_t \Lambda_{t,t+1} \Pi_{t+1}^{-1} \left[1 + \kappa_{s,t+1}^f Q_{s,t+1} M_{2,s,t+1} \right] \tag{30}$$

$$\frac{M_{1,s,t} - 1}{M_{2,s,t} - 1} = \psi_s \tag{31}$$

The variable p_t^k is the relative price of new capital (Tobin's Q); $w_{s,t}$ is the real wage; and $p_{m,s,t}$ is the relative price of the wholesale output in segment s. Equation 27 is the standard static rule that firms use to decide how much labor to hire. Equation 28 governs the optimal utilization rate: the firm equates the marginal cost of higher utilization (faster depreciation, valued at the shadow price of capital $p_t^k M_{1,s,t}$) to the marginal product of utilization. Both $M_{1,s,t}$ and $M_{2,s,t}$ originate from the same loan-in-advance constraint (Equation 23) that forces firms to issue bonds to finance their investments. Equations 29 and 30 represent the optimal choices for holding capital and bonds, respectively. As evident from the Lagrangian equation (Equation 26), both wedges are functions of the single multiplier $\nu_{2,s,t}$ associated with this financing constraint. If the constraint is not actually binding, then both $M_{1,s,t}$ and $M_{2,s,t}$ would simply equal one, and these equations would appear similar to the standard rules for asset pricing. Because this constraint is binding, however, $M_{1,s,t}$ acts as an "investment wedge" and $M_{2,s,t}$ as a "financial wedge." These wedges distort the firm's standard decisions, and the central idea is that fluctuations in these wedges are the primary way in which unconventional monetary policies such as QE/QT are transmitted to the real economy.

4.5 Monetary Policy

4.5.1 Conventional monetary policy

Before discussing unconventional monetary policy, we must first define conventional policy. We define this as the central bank's adjustment of the short-term interest rate

R_t^{tr}. This adjustment is described by an internal feedback rule, similar to the one proposed by Taylor (1993):

$$\begin{aligned}\log R_t^{tr} = (1-\rho_r)\log R^{tr} + \rho_r \log R_{t-1}^{tr} \\ + (1-\rho_r)\left[\phi_\pi(\log \Pi_t - \log \Pi) + \phi_y(\log Y_t - \log Y_{t-1})\right] + s_r \varepsilon_{r,t}\end{aligned} \tag{32}$$

In this rule, R^{tr} and Π represent the long-run "steady-state" values for the policy rate and the inflation target. The parameters $0 < \rho_r < 1$, ϕ_π, and ϕ_y are all non-negative numbers. To ensure the model has a stable solution (a "determinate equilibrium"), we only consider cases where $\phi_\pi > 1$. This rule simply means that the policy rate adjusts whenever inflation moves away from its target, or when output growth moves away from its trend (which we assume is zero in this model). We also assume that the central bank sets the interest rate on reserves the same as the main policy rate. Therefore, the deposit rate (R_t^d) and the reserve rate (R_t^{re}) are both equal to R_t^{tr}:

$$R_t^d = R_t^{re} = R_t^{tr} \tag{33}$$

4.5.2 Unconventional monetary policy

Quantitative easing is arguably the most significant unconventional monetary policy employed by central banks. It was first adopted by the Bank of Japan in the early 2000s. After the Great Recession, major economies such as the United States, the euro area, and the United Kingdom implemented this tool, but its use expanded to an unprecedented scale in response to the 2020 COVID-19 pandemic.

While the Federal Reserve's balance sheet reached \$4.5 trillion after its initial post-crisis programs, it surged to a peak of nearly \$9 trillion (around 36% of U.S. GDP) by early 2022. The European Central Bank's balance sheet, which stood at €4.7 trillion in late 2018, peaked at €6.89 trillion (approximately 50% of euro area GDP) by the end of 2023. The Bank of Japan's holdings, which long exceeded 100% of its GDP, grew to a high of over 764 trillion JPY.

This era of massive expansion has now begun to pivot. Since 2024, these central banks have entered a new phase of QT, or QN, as the ECB terms it. This shift involves discontinuing reinvestments, such as the ECB's full halt of its APP and PEPP programs by the end of 2024, and actively allowing these massive balance sheets to shrink further.

Following Gertler and Karadi (2011), Gertler and Karadi (2013), and Carlstrom et al. (2017), we define quantitative easing as a central bank's purchase of long-term government bonds. These purchases are made by creating new interest-bearing reserves held by the financial intermediaries. In our model, the central bank's balance sheet is:

$$Q_t F_{cb,t} + Q_{B,t} B_{cb,t} = \mathrm{RE}_t \tag{34}$$

The central bank holds long-term government bonds ($B_{cb,t}$) and long-term private bonds ($F_{cb,t}$) as assets financed by issuing interest-bearing reserves (RE_t). Any profit (operating surplus) from these holdings is transferred to the government fiscal authority. The model's market-clearing condition requires that all bonds from the government are held by either financial intermediaries or the central bank. QE/QT policies can have real effects on the economy, but only if financial intermediaries are constrained by the costly enforcement problem. When this constraint is active (or "binds"), the central bank's bond purchases (financed by new reserves) help to ease this constraint. In this situation, the central bank's demand for bonds adds to, rather than "crowds out," the intermediaries' demand. This increases total demand for bonds, leading to higher bond prices. Higher bond prices relax the loan-in-advance constraint faced by wholesale goods producers. This ultimately results in higher investment and greater aggregate demand. However, if the intermediaries' constraint is not binding, or if wholesale firms do not need to borrow to finance investment (i.e., $\psi = 0$), then QE/QT has no economic effects. Furthermore, assuming $\Delta < 1$ and that both constraints are binding, central bank purchases of long-term private bonds have a stronger impact on excess returns than purchases of long-term government bonds, making them a more powerful stimulus. We treat QE/QT as an exogenous policy.

We assume that the central bank's bond holdings follow an external AR(1) process:

$$b_{cb,t} = (1 - \rho_b) b_{cb} + \rho_b b_{cb,t-1} + s_b \varepsilon_{b,t} \tag{35}$$

$$f_{cb,t} = (1 - \rho_f) f_{cb} + \rho_f f_{cb,t-1} + s_f \varepsilon_{f,t} \tag{36}$$

Here, b_{cb} and f_{cb} are the steady-state (long-run) levels of real central bank government and private bond holdings, ρ_b and ρ_f are persistence parameters (between 0 and 1), and $\varepsilon_{b,t}$ and $\varepsilon_{f,t}$ are stochastic shocks with standard deviations s_b and s_f.

4.6 Calibration

The model is solved using a linear approximation around the non-stochastic steady state and calibrated at a quarterly frequency. A set of standard structural parameters—including household preferences, production technology, and aggregate fiscal and monetary policy rules—are set to conventional values found in the New Keynesian literature. For brevity, these common aggregate parameters are detailed in Table 5 in Appendix C.

Our primary focus is on the segment-specific heterogeneity and financial frictions. The calibrated values for these parameters are listed in Table 4. The economy is composed of two distinct segments, labeled "Low" (L) and "High" (H), characterized by asymmetric balance sheet constraints and maturity structures. The Low segment represents the dominant share of the economy ($\zeta_L = 0.90$), while the High segment represents the remainder ($\zeta_H = 0.10$). To discipline the financial intermediaries, we directly target their maturity structures, deposit ratios, and steady-state spreads, allowing the model to endogenously determine the remaining financial ratios.

Specifically, we calibrate the target maturity of private loans to 12 quarters for the Low segment and 36 quarters for the High segment. The maturities of government bonds, central bank reserves, and deposits are set to match standard empirical counterparts. Reflecting their distinct risk and duration profiles, we assign different steady-state net spreads: 200 basis points for the Low segment and 400 basis points for the High segment.

Driven by these targeted maturities and spreads, the model endogenously generates the distinct balance sheet characteristics of each segment. For the Low segment, this calibration implies a steady-state leverage ratio (Lev_L) of 2.22, a liquidity coverage ratio (θ_L) of 0.75, and a recoverability parameter (Δ_L) of 0.50. In contrast, the longer maturity structure and higher spread in the High segment endogenously produce a higher leverage ratio of 4.00, tighter liquidity buffers ($\theta_H = 0.32$), and a lower recoverability parameter ($\Delta_H = 0.25$).

Table 4: Calibrated Parameters: Segment-Specific (Low vs. High)

		Segment Value	
Param.	**Description**	**Low (L)**	**High (H)**
Financial Frictions & Balance Sheet			
ζ_s	Share of the segment	0.90	0.10
Lev_s	Leverage ratio	2.22	4.00
θ_s	Liquidity coverage ratio	0.75	0.32
Dep_s	Target deposit ratio	0.55	0.75
Δ_s	Recoverability parameter	0.50	0.25
Sp_s^f	Steady-state net spread	200 bps	400 bps
Maturity Structure			
M_s^f	Target loan maturity (Qtrs)	12	36
M^b	Target government bond maturity (Qtrs)	40	40
M^{re}	Target reserve maturity (Qtrs)	1	1
M^d	Target deposit maturity (Qtrs)	1	1
Segment-Specific Shocks			
ρ_A, s_A	Productivity shock	0.95, 0.01	
ρ_θ, s_θ	Liquidity shock	0.95, 0.01	
ρ_f, s_f	CB private loan holdings	0.80, 0.10	
$\rho_{\kappa^f}, s_{\kappa^f}$	Duration of private loans shock	0.95, 0.01	
ρ_D, s_D	Recoverability shock	0.95, 0.00	

Note: Parameters with a single value are calibrated symmetrically across segments. Spreads are expressed in annualized basis points (bps). Shock volatilities set to zero indicate that the corresponding shock is inactive in the baseline calibration.

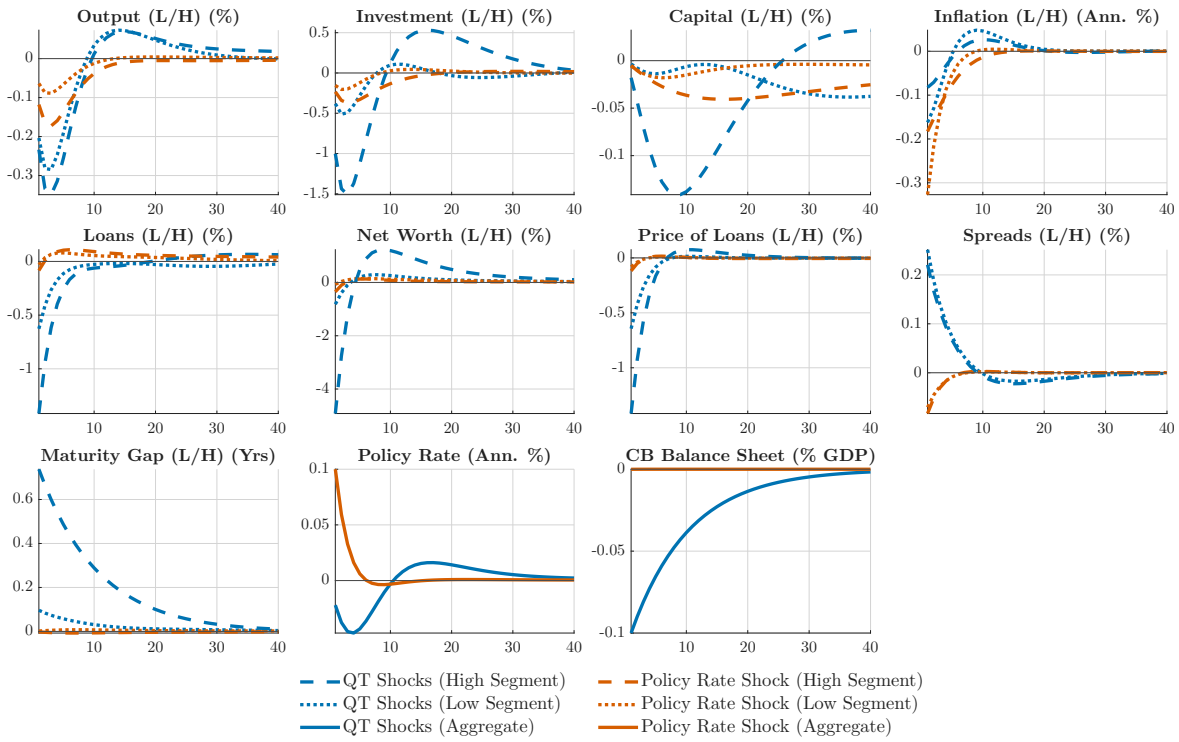
5 Discussion

In this section, we compare the effects of exogenous shocks on both conventional and unconventional policy tools. We compare two types of shocks: (i) a conventional monetary policy tightening shock (i.e., an unexpected increase in the desired policy rate, $\varepsilon_{r,t}$), and (ii) a QT shock (i.e., an unexpected contraction in the central bank's government bond holdings, $\varepsilon_{b,t}$). Subsequently, we examine the effects of liquidity and technology shocks.

5.1 Conventional (Policy Rate) and Unconventional (QT) Shocks

Figure 5 isolates the impact of bank heterogeneity by plotting the segment-level responses to conventional (policy rate) and unconventional (QT) shocks. This differential analysis provides a visual test of how the financial structure—specifically the maturity

Figure 5: Responses of High vs. Low Maturity Gap Segments to Policy Rate and QT Shocks

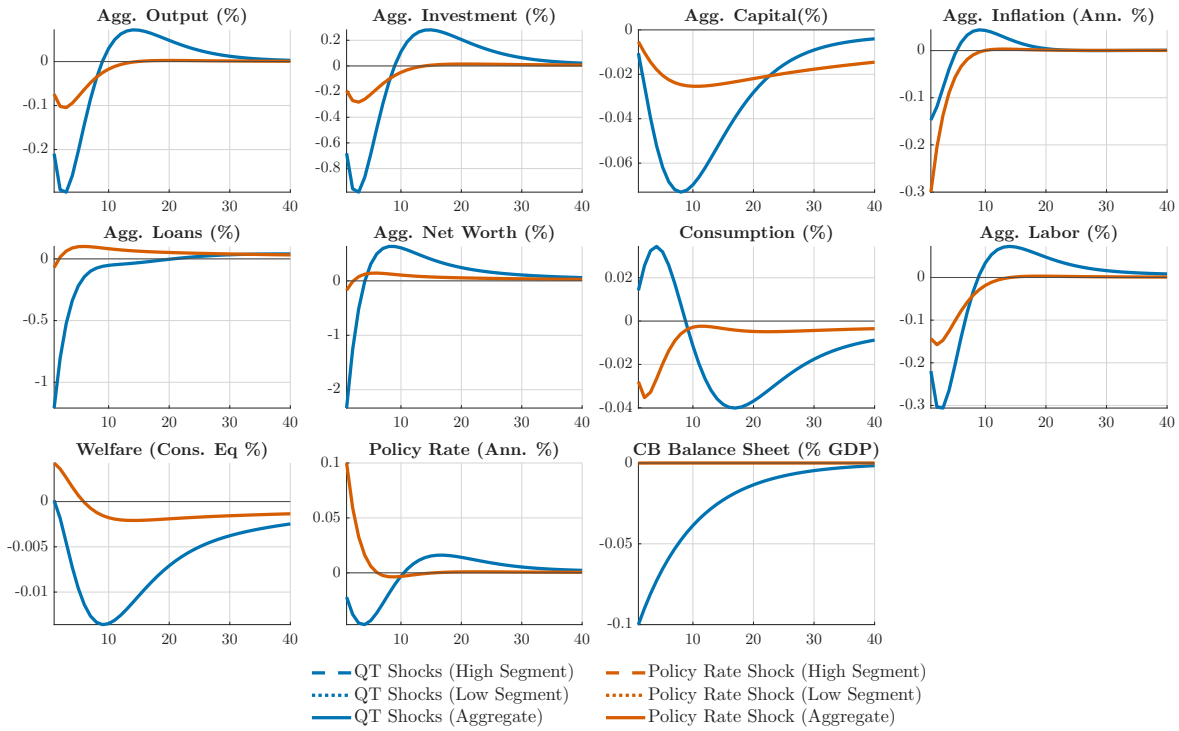


mismatch—mediates different policy tools.

The transmission mechanisms of these shocks are fundamentally different, which explains the varied outcomes observed in Figure 5. The conventional policy rate hike (orange lines) operates primarily through the intertemporal substitution channel and by raising the short-term funding cost for banks. As the central bank raises its policy rate, deposit rates rise uniformly. This squeezes net interest margins similarly across all banks, regardless of their asset duration. Consequently, the segment-level responses are virtually identical; the orange High- and Low-segment lines track each other closely across all variables, confirming that the maturity gap is almost irrelevant for the transmission of conventional policy.

In sharp contrast, the unconventional QT shock (blue lines) operates directly through the bank balance sheet channel, which is highly sensitive to maturity mismatches. QT involves the central bank reducing its government bond holdings, which drains reserves and increases the supply of long-term assets in the market. Banks holding long-duration assets funded by short-duration liabilities suffer mark-to-market valuation losses due

Figure 6: Aggregate Responses to Policy Rate and QT Shocks



to the resulting spike in yields.

This mechanism is clearly visible in the Net Worth and Price of Loans panels of Figure 5. Under the QT shock, High-segment (H) banks suffer a severe capital hit—net worth drops by approximately 4%—creating a large divergence compared to the Low segment (L), where the decline is roughly half as large. The corresponding fall in the price of loans reaches approximately 1% in the High segment. This drop in net worth causes the leverage constraint to bind more tightly. To restore their capital ratios, High-segment banks are forced to deleverage aggressively. The Spreads panel shows a sharp widening of the credit spread—by approximately 0.2 percentage points in the High segment—reflecting the scarcity of bank capital. This "financial wedge" forces a contraction in credit supply, visible in the Loans panel, where the High segment experiences a decline in lending of approximately 1%, compared to a much smaller decline in the Low segment. The Maturity Gap panel reveals that the QT shock widens the effective maturity gap by approximately 0.6 years in the High segment, further illustrating how this shock interacts directly with the duration structure of

bank balance sheets.

Figure 6 demonstrates how this heterogeneity translates into aggregate macroeconomic outcomes. Both policy actions successfully trigger a contraction, causing aggregate output, investment, and consumption to fall. However, the unconventional QT shock is substantially more powerful, and its effects are far more persistent. Under the QT shock, aggregate output drops by approximately 0.3%—roughly three times the decline caused by the conventional policy rate shock—while aggregate investment contracts by approximately 1%. The QT shock also produces a notable deflationary impulse, with aggregate inflation declining by approximately 0.15 percentage points (annualized), in contrast to the near-zero inflationary response under the policy rate shock. This deflationary effect arises because the severe credit contraction depresses aggregate demand by more than the rate hike alone.

The Aggregate Investment panel highlights the amplification through the balance sheet channel. Under the QT scenario, the credit crunch described above starves firms of capital. Investment contracts by roughly twice the amount observed under the conventional policy shock. Aggregate net worth falls by approximately 2%, driving a sustained credit contraction (aggregate loans decline by approximately 1%) that produces a deeper and longer-lasting fall in aggregate output.

Crucially, the Welfare panel summarizes the asymmetric costs of these policies. The QT shock induces a welfare loss of approximately 0.015% in consumption-equivalent terms—substantially larger and more persistent than the welfare cost of the conventional rate shock.

Interestingly, the Policy Rate panel reveals a further asymmetry. While the conventional policy rate shock directly raises the short-term rate and then allows it to revert, the QT shock triggers a sharp endogenous increase in the policy rate—reaching approximately 0.1 percentage points (annualized)—as the Taylor rule responds to the macroeconomic fallout. This means that, in practice, QT generates an additional monetary tightening through its aggregate effects, compounding the initial balance sheet shock with a subsequent rise in funding costs.

This analysis confirms that maturity mismatch functions as a potent systemic amplifier for unconventional shocks. While conventional rate changes act as a broad tool affecting intertemporal decisions uniformly across the banking system, QT operates directly through the bank balance sheet channel, where it is amplified by maturity mismatches. When the banking sector is characterized by high maturity transfor-

mation, QT triggers a severe credit contraction that extends well beyond the initial portfolio adjustment.

5.2 Responses to Real and Financial Shocks

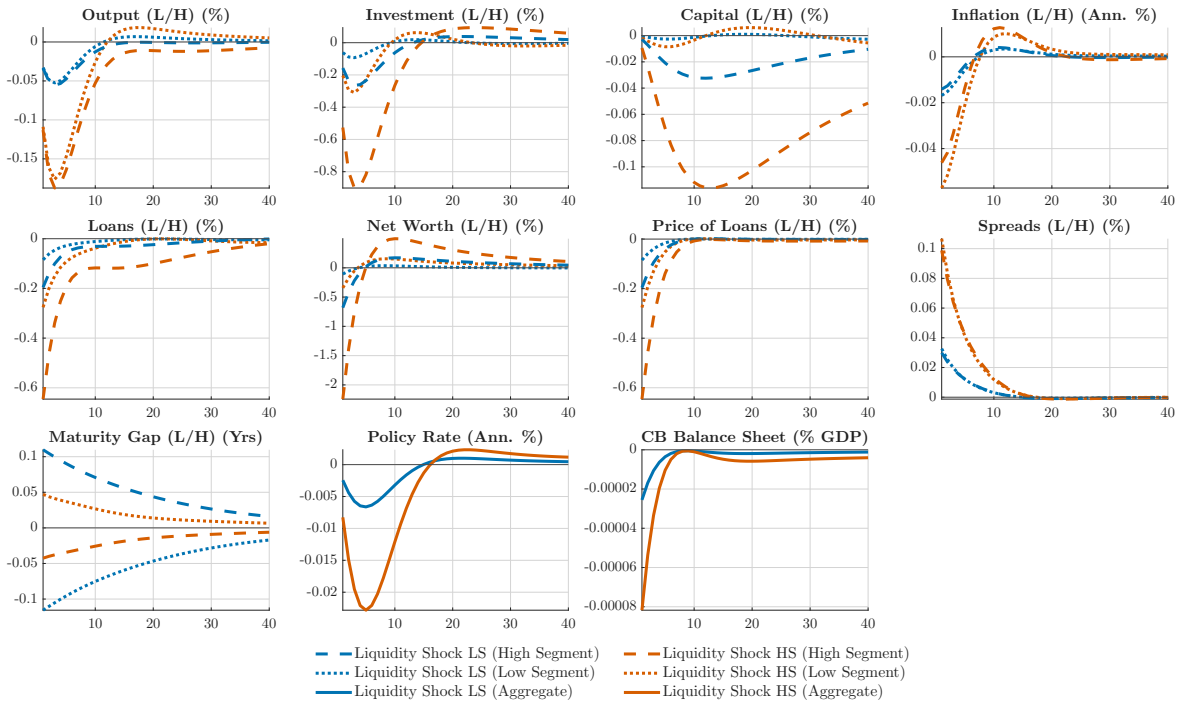
Having established that bank maturity mismatch is a critical determinant for the transmission of unconventional monetary policy but not conventional policy, we now broaden the analysis. We examine two non-policy shocks to understand the wider implications of this financial structure for the economy. First, we introduce a liquidity shock (a pure financial friction shock) to test whether the high-maturity-gap structure serves as a more general source of systemic fragility, amplifying crises that originate within the financial system itself, independent of any policy action. Second, we introduce a positive technology shock (a real, supply-side shock) to test whether the banking system's maturity gap also mediates the economy's ability to capitalize on positive opportunities or whether it acts as a structural drag on growth.

5.2.1 Impulse Response to Liquidity Shock

Figures 7 and 8 illustrate the economy's response to a tightening of bank constraints, modeled as a sudden exogenous positive shock to $\theta_{s,t}$ (a "liquidity shock"). We analyze two distinct scenarios based on the segment from which this financial disturbance originates: a shock originating in the Low segment (L), characterized by a low maturity mismatch, versus a shock originating in the High segment (H), characterized by a high maturity mismatch.

This shock acts as a sudden tightening of funding conditions, generating a clear macroeconomic contraction. The aggregate results (Figure 8) demonstrate that financial frictions can drive a real-sector downturn: aggregate output, investment, and consumption all decline. The transmission mechanism works through the banking sector's balance sheet. The shock forces banks to deleverage to satisfy the tighter constraint. This is visible in the contraction of aggregate loans, which creates a credit crunch. Because firms rely on these loans for working capital (via the loan-in-advance constraint) and capital accumulation, the reduction in credit availability directly affects the real economy. Investment contracts sharply, and the economy enters a recession. In response to the resulting deflationary pressure, the central bank cuts the policy rate to stimulate demand.

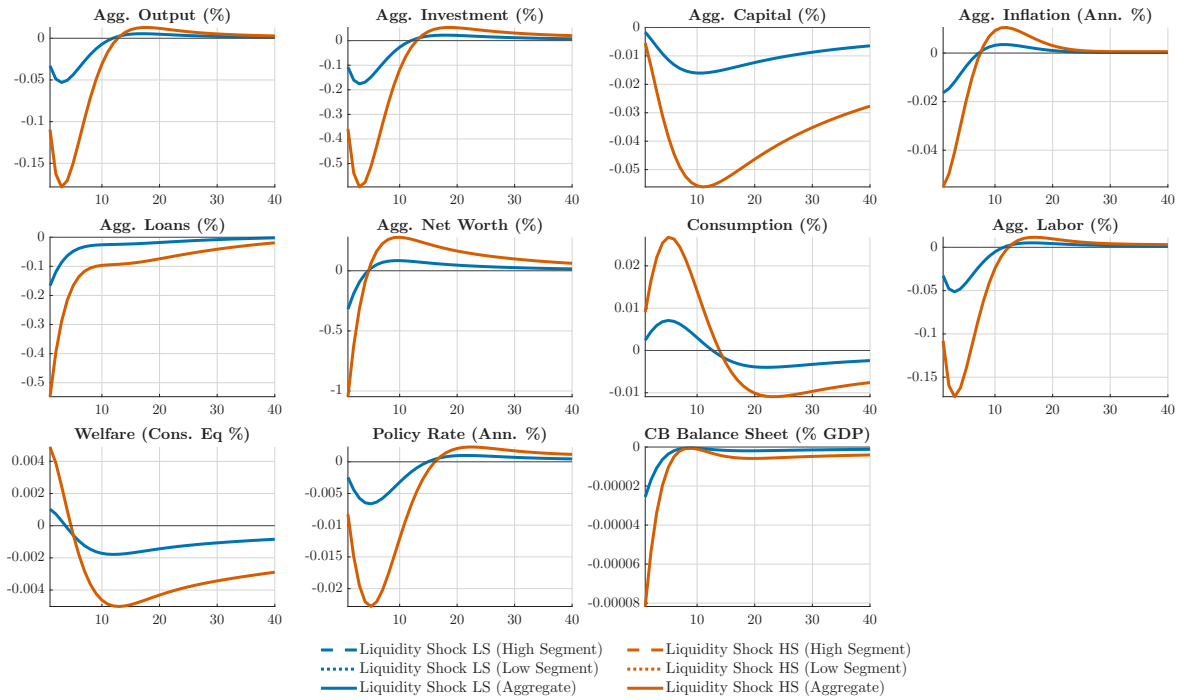
Figure 7: Segment-Level Responses to Liquidity Shock



However, the quantitative impact depends heavily on the segment from which the shock originates. A shock hitting the High segment triggers a far more severe crisis than one hitting the Low segment. When the liquidity shock originates in the High segment, the contraction is amplified substantially. As shown in Figure 8, aggregate output falls by approximately 0.15%, and aggregate investment declines by approximately 0.5%—impacts that are several times larger than those observed in the Low-segment scenario. Aggregate net worth falls by approximately 1%, driving a corresponding contraction in aggregate loans of approximately 0.5%. This deeper recession is driven by the structural vulnerability inherent in the High segment's balance sheet. Due to its high degree of maturity transformation, the High segment is heavily dependent on rolling over short-term funding to finance long-term assets. Consequently, when funding conditions tighten, the shock forces aggressive deleveraging.

Figure 7 highlights the heterogeneity driving this divergence. At the segment level, the High-segment shock produces markedly larger responses than the Low-segment shock across all financial variables. The Price of Loans panel shows a decline of approximately 0.6% in the High segment, while the Spreads panel indicates a rise of

Figure 8: Aggregate Responses to Liquidity Shock

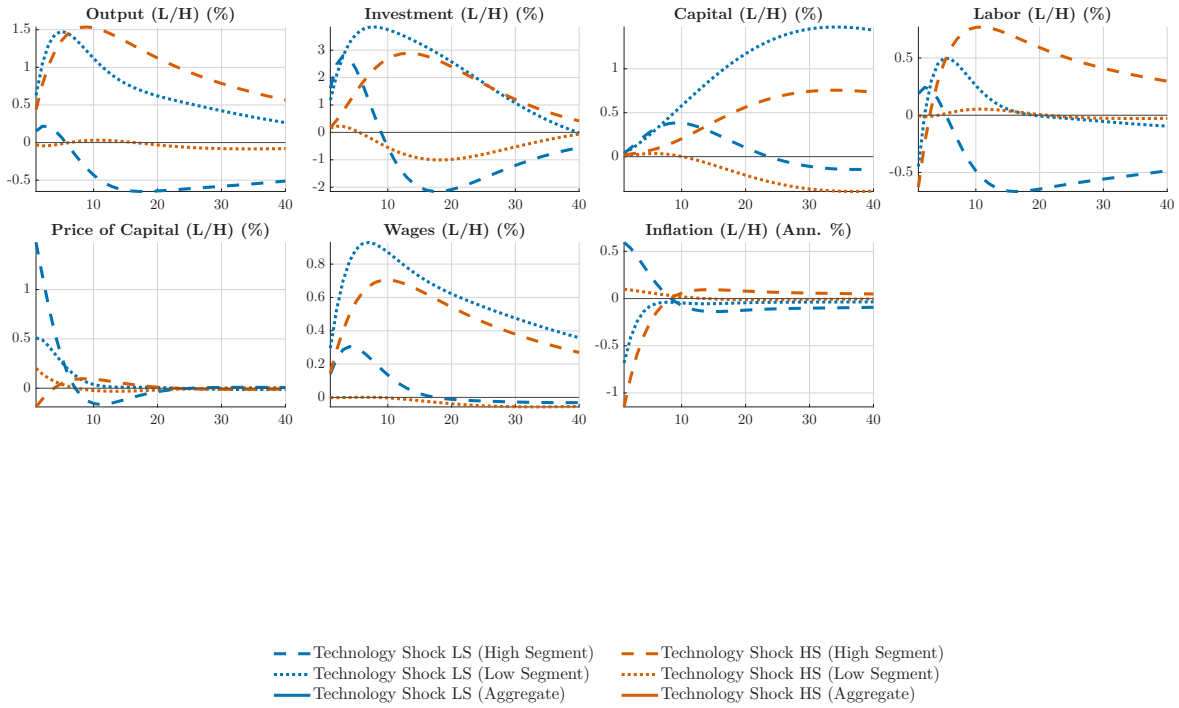


approximately 0.1 percentage points—both substantially larger than the corresponding Low-segment responses. This relative tightening of credit conditions implies that the financial wedge widens significantly more when the shock originates in the High segment. The Investment panel further confirms this, showing that segment-level investment drops by approximately 0.8% in the High segment under the HS shock—roughly four times larger than under the LS shock. The Net Worth panel reveals the underlying driver: High-segment net worth declines by approximately 1.5–2%, reflecting the acute vulnerability of maturity-mismatched balance sheets to funding shocks.

In summary, the High segment's financial structure acts as a magnifier for financial shocks. Because of its reliance on maturity transformation, a liquidity squeeze in this segment translates into a sharp spike in funding costs and a contraction in lending, rendering the central bank's accommodative policy less effective. The intended stimulus from the rate cut is overwhelmed by the widening credit spreads, leading to deeper welfare losses and economic contraction.

5.2.2 Impulse Response to Technology Shock

Figure 9: Segment-Level Responses to Technology Shock

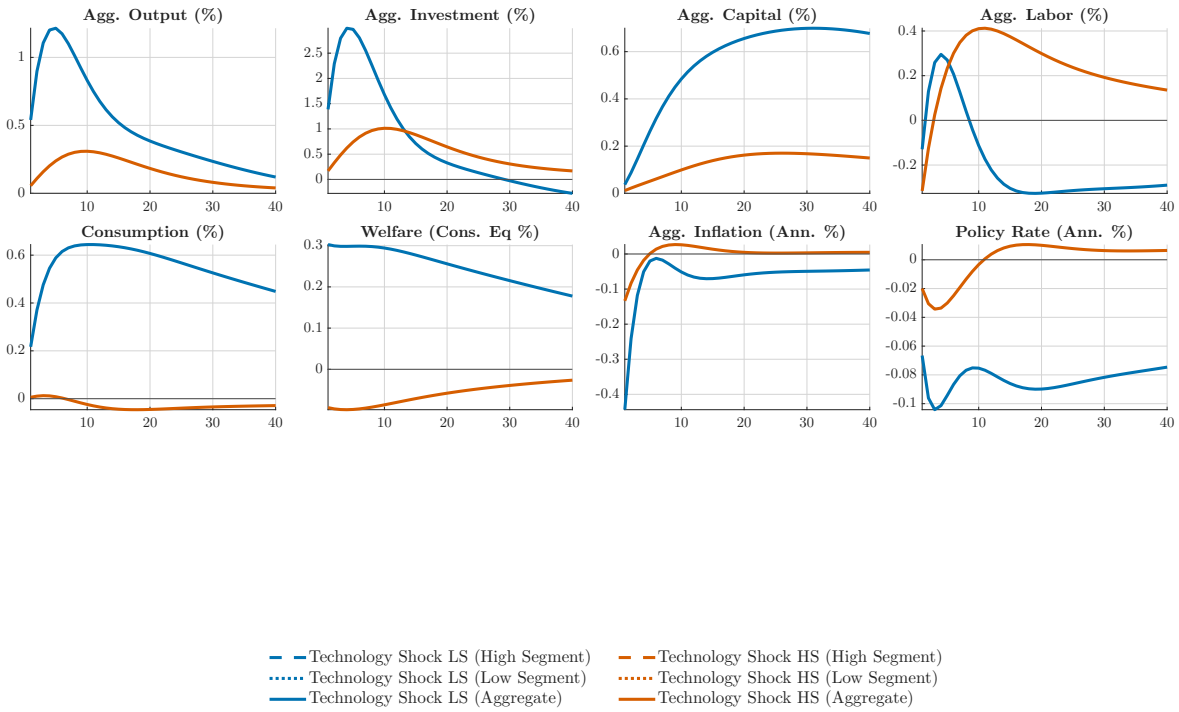


Figures 9 and 10 analyze the economy's response to a positive technology shock. We distinguish between two scenarios based on the segment in which the productivity gain originates: a shock in the Low segment (L), characterized by a low maturity mismatch, versus a shock in the High segment (H), characterized by a high maturity mismatch.

The aggregate results (Figure 10) align with standard macroeconomic theory: both shocks lead to a persistent expansion, with aggregate output, investment, and consumption all rising. As the economy's productive capacity expands and outpaces aggregate demand, downward pressure on prices causes inflation to decrease—by approximately 0.4 percentage points (annualized) in the Low-segment scenario and approximately 0.2 percentage points in the High-segment scenario. This benign disinflationary environment gives the central bank (via the Taylor rule) clear justification to adopt an accommodative stance, as seen in the decline of the policy rate to approximately -0.1% (annualized) in the Low-segment scenario.

However, the quantitative differences between the two scenarios highlight that the

Figure 10: Aggregate Responses to Technology Shock



banking system's structure plays a critical role in mediating this positive shock. The Low-segment shock triggers a significantly stronger economic expansion compared to the High-segment shock. As shown in Figure 10, aggregate investment in the Low-segment scenario surges to a peak of approximately 2.5%, more than double the response observed when the shock hits the High segment (approximately 1%). Aggregate output rises by approximately 1% under the Low-segment shock, compared to approximately 0.5% under the High-segment shock. Consequently, the Low-segment shock leads to a higher trajectory for consumption (peaking at approximately 0.6% versus 0.3%) and a more significant improvement in welfare (approximately 0.3% versus 0.15% in consumption-equivalent terms).

The muted response in the High segment is explained by the structural friction inherent in its balance sheet. While the technology shock increases profitable investment opportunities, the High segment's reliance on maturity transformation creates a bottleneck. Figure 9 illustrates this mechanism at the segment level. The Investment panel shows that the Low-segment shock catalyzes a large reallocation of resources, with Low-segment investment surging by approximately 3%, while the High-segment

shock generates a response of approximately 1%. The Capital and Price of Capital panels confirm that this investment differential translates into a substantial divergence in capital accumulation and asset prices: under the Low-segment shock, the price of capital rises by approximately 1%, whereas the High-segment shock produces a notably weaker response.

Crucially, the Wages panel reveals that the Low-segment shock generates a sharp, immediate spike in the relative wage of approximately 0.8%, sending a strong price signal that efficiently reallocates labor to the more productive segment. The Labor panel corroborates this, showing a corresponding reallocation of employment. In contrast, the High-segment shock fails to generate this strong market signal. Because the High-segment banks are constrained by their maturity mismatch, they cannot expand credit as efficiently to fund the new investment opportunities. This financial friction acts as a structural drag, dampening the pass-through of the accommodative policy rate and limiting the economy's capacity to capitalize on the productivity gains.

In summary, this analysis demonstrates that the banking system's maturity structure is not just a vulnerability that amplifies negative shocks (as seen in the liquidity crisis); it is also a limiting factor during positive expansions. A less flexible, high-maturity-gap banking segment acts as a brake on growth, impeding the efficient allocation of new capital even in the face of positive fundamental shocks.

6 Concluding Remarks

This study demonstrates that the financial system's structure—specifically, the degree of maturity transformation within the banking sector—is a critical determinant of macroeconomic outcomes and the propagation of various shocks. This conclusion is most evident when comparing the effects of different monetary policy tools.

We find a "shock-specific" asymmetry between conventional (policy rate) and unconventional (QT) policies. Conventional policy rate hikes operate broadly through intertemporal substitution and rising funding costs, and their effects are largely indifferent to banks' maturity gaps. In contrast, unconventional policy (QT) operates directly through the bank balance sheet channel. By inflicting valuation losses on long-duration assets, QT severely tightens the leverage constraints of banks with high maturity gaps. Thus, this financial structure functions as a potent amplifier of unconventional policies, leading to a deeper and more persistent economic downturn and significantly larger welfare losses.

The effects of banking structure extend beyond monetary policy. We find that a high-maturity-gap structure not only amplifies negative shocks but also dampens positive ones. Such a structure renders the banking segment substantially more vulnerable to negative financial shocks. Conversely, in response to positive technology shocks, the high-maturity-gap banking segment acts as a bottleneck. By failing to efficiently reallocate capital to productive opportunities, it constrains credit expansion and mutes the potential economic boom.

By connecting detailed supervisory data with a structural model of banking behavior, this study highlights *the central role of maturity transformation* in monetary transmission and in overall financial and economic stability. Our combined theoretical and empirical findings have significant policy implications. First, they suggest that the choice between policy tools (e.g., rate hikes vs. QE/QT) is *not neutral* and must account for the prevailing financial structure and potential welfare costs. Second, the results provide a strong justification for macroprudential policies aimed at limiting excessive maturity transformation. Thus, monitoring and managing maturity mismatches is essential not only for prudential oversight but also for enhancing macroeconomic resilience, reducing systemic risk, and understanding how different policy tools and macroeconomic disturbances propagate through the banking system and the broader economy.

References

Allen, F. and Gale, D. (1994). Limited market participation and volatility of asset prices. *The American Economic Review*, 84(4):933–955.

Allen, F. and Gale, D. (2007). *Understanding Financial Crises*. Oxford University Press, Oxford.

Altavilla, C., Brugnolini, L., Gürkaynak, R. S., Motto, R., and Ragusa, G. (2019). Measuring euro area monetary policy. *Journal of Monetary Economics*, 108:162–179.

Bernanke, B. S., Gertler, M., and Gilchrist, S. (1999). The financial accelerator in a quantitative business cycle framework. In *Handbook of Macroeconomics*, volume 1, pages 1341–1393. Elsevier.

Carlstrom, C. T., Fuerst, T. S., and Paustian, M. (2017). Targeting long rates in a model with segmented markets. *American Economic Journal: Macroeconomics*, 9(1):205–242.

Christiano, L. J., Eichenbaum, M., and Evans, C. L. (2005). Nominal rigidities and the dynamic effects of a shock to monetary policy. *Journal of Political Economy*, 113(1):1–45.

Coulier, L., Pancaro, C., and Reghezza, A. (2024). Are low interest rates firing back? interest rate risk in the banking book and bank lending in a rising interest rate environment. *ECB Working Paper Series*.

Di Tella, S. and Kurlat, P. (2021). Why are banks exposed to monetary policy? *American Economic Journal: Macroeconomics*, 13(4):295–340.

Diamond, D. W. and Dybvig, P. H. (1983). Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91(3):401–419.

Drechsler, I., Savov, A., and Schnabl, P. (2017). The deposits channel of monetary policy. *The Quarterly Journal of Economics*, 132(4):1819–1876.

English, W. B., Van den Heuvel, S. J., and Zakrajšek, E. (2018). Interest rate risk and bank equity valuations. *Journal of Monetary Economics*, 98:80–97.

Esposito, L., Nobili, A., and Ropele, T. (2015). The management of interest rate risk during the crisis: Evidence from italian banks. *Journal of Banking & Finance*, 59:486–504.

Flannery, M. J. and James, C. M. (1984). The effect of interest rate changes on the common stock returns of financial institutions. *The Journal of Finance*, 39(4):1141–1153.

Garofalo, M. (2024). Granular monetary policy. Working Paper.

Gertler, M. and Karadi, P. (2011). A model of unconventional monetary policy. *Journal of Monetary Economics*, 58(1):17–34. Carnegie-Rochester Conference Series on Public Policy: The Future of Central Banking, April 16–17, 2010.

Gertler, M. and Karadi, P. (2013). Qe 1 vs. 2 vs. 3: A framework for analyzing large-scale asset purchases as a monetary policy tool. *International Journal of Central Banking*, 9(S1):5–53.

Gertler, M. and Kiyotaki, N. (2010a). Chapter 11 - financial intermediation and credit policy in business cycle analysis. volume 3 of *Handbook of Monetary Economics*, pages 547–599. Elsevier.

Gertler, M. and Kiyotaki, N. (2010b). Financial intermediation and credit policy in business cycle analysis. In *Handbook of Monetary Economics*, volume 3, pages 547–599. Elsevier.

Ghassibe, M. (2021). Monetary policy and production networks: an empirical investigation. *Journal of Monetary Economics*, 119:21–39.

Gomez, M., Landier, A., Sraer, D., and Thesmar, D. (2021). Bank exposure to interest-rate risk and the transmission of monetary policy. Technical report, NBER Working Paper No. 28551.

Heider, F., Saidi, F., and Schepens, G. (2019). Life below zero: Bank lending under negative policy rates. *The Review of Financial Studies*, 32(10):3728–3761.

Jordà, O. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.

Macaulay, F. R. (1938). Some theoretical problems suggested by the movements of interest rates, bond yields and stock prices in the united states since 1856. *NBER Books*.

Nakamura, E., Riblier, V., and Steinsson, J. (2025). Beyond the taylor rule. NBER Working Paper 34200, National Bureau of Economic Research.

Paul, P. (2022). Interest rate risk and bank net interest margins. *Review of Financial Studies*, 35(11):5041–5084.

Paul, P. (2023). Banks, maturity transformation, and monetary policy. *Journal of Financial Intermediation*, 53:101011.

Purnanandam, A. (2007). Interest rate derivatives at commercial banks: An empirical investigation. *Journal of Monetary Economics*, 54(6):1769–1808.

Schnabel, I. (2025). Towards a new Eurosystem balance sheet. Speech at the ECB Conference on Money Markets 2025.

Sims, E. and Wu, J. C. (2021). Evaluating central banks' tool kit: Past, present, and future. *Journal of Monetary Economics*, 118:135–160.

Smets, F. and Wouters, R. (2007). Shocks and frictions in US business cycles: A Bayesian DSGE approach. *American Economic Review*, 97(3):586–606.

Taylor, J. B. (1993). Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39(1):195–214.

Van den Heuvel, S. J. (2002). Does bank capital matter for monetary transmission? *Economic Policy Review*, 8(1):259–265.

Varraso, G. (2024). Bank interest rate risk exposure and monetary policy amplification. Working Paper.

Wang, X. (2023). A macro-financial perspective to analyse maturity mismatch and default. *Journal of Banking and Finance*, 151:106468.

Woodford, M. (2001). Fiscal requirements for price stability. *Journal of Money, Credit and Banking*, 33(3):669–728.

Appendix A Full Model and Derivations

A.1 Households

The model contains a large population of identical households. We can analyze the behavior of a single representative household because all households behave similarly. Our framework is based on Gertler and Karadi (2013). Each household consists of two types of members: workers and intermediaries. The proportions of these two types are fixed. There are two intermediary segments indexed by $s \in \{L, H\}$ (low vs. high). A type-s intermediary survives into the next period with probability σ_s; with probability $1 - \sigma_s$ it exits and is replaced by a worker who becomes a new type-s intermediary endowed with start-up net worth X_s.

The lifetime utility of the representative household is

$$E_t \sum_{j=0}^{\infty} \beta^j \left(\psi^c_{t+j} \ln(C_{t+j} - bC_{t+j-1}) - \psi^c_{t+j}\psi^l_{t+j} \left[\chi_L \frac{L^{1+\eta}_{L,t+j}}{1+\eta} + \chi_H \frac{L^{1+\eta}_{H,t+j}}{1+\eta} \right] \right), \tag{37}$$

where β is the discount factor, b is internal habit formation, χ_s scales disutility of labor in sector s, and η is the inverse Frisch elasticity. ψ^c_t is an intertemporal preference shifter for consumption and ψ^l_t is an intratemporal shifter for labor. The household supplies labor to each sector, $(L_{L,t}, L_{H,t})$, and consumes the final composite good C_t.

Households face the following nominal budget constraint:

$$\begin{aligned} P_t C_t + D_t - D_{t-1} \leq\ & MRS_{L,t} L_{L,t} + MRS_{H,t} L_{H,t} + DIV_t \\ & - P_t(X_L + X_H) - P_t T_t \\ & + \sum_{s \in \{L,H\}} (R^d_{s,t-1} - 1) D_{s,t-1}, \end{aligned} \tag{38}$$

where P_t is the final-goods price, $D_{s,t}$ are deposits placed in the two deposit blocks paying gross rates $R^d_{s,t}$, and D_t is the deposit composite (defined below). $MRS_{s,t}$ denotes the nominal payment the household receives for supplying labor in sector s (paid via unions), and $mrs_{s,t} \equiv MRS_{s,t}/P_t$ is the corresponding real payment. DIV_t is household dividend income (including returns from firms and exiting intermediaries), X_s are start-up transfers, and T_t are lump-sum taxes.

Deposit aggregation. The deposit composite (in real terms, $d_{s,t} \equiv D_{s,t}/P_t$ and $d_t \equiv D_t/P_t$) is CES:

$$d_t = \left[\omega_d \, d_{L,t}^{\frac{\eta_d - 1}{\eta_d}} + (1-\omega_d) \, d_{H,t}^{\frac{\eta_d - 1}{\eta_d}}\right]^{\frac{\eta_d}{\eta_d - 1}}, \tag{39}$$

with elasticity $\eta_d > 0$ and weight $\omega_d \in (0,1)$. The dual (gross) deposit rate aggregator is

$$(R_t^d)^{1-\eta_d} = \omega_d (R_{L,t}^d)^{1-\eta_d} + (1-\omega_d)(R_{H,t}^d)^{1-\eta_d}. \tag{40}$$

Optimal allocation across deposit blocks satisfies

$$\frac{d_{L,t}}{d_{H,t}} = \left(\frac{\omega_d}{1-\omega_d}\right)^{\eta_d} \left(\frac{R_{L,t}^d}{R_{H,t}^d}\right)^{\eta_d}. \tag{41}$$

First-order conditions. The first-order conditions for the household are

$$\mu_t = \frac{\psi_t^c}{C_t - bC_{t-1}} - b\beta E_t \frac{\psi_{t+1}^c}{C_{t+1} - bC_t}, \tag{42}$$

$$\Lambda_{t,t+1} = \beta \frac{\mu_{t+1}}{\mu_t}, \tag{43}$$

$$\psi_t^c \psi_t^l \chi_s L_{s,t}^\eta = \mu_t \, mrs_{s,t}, \qquad s \in \{L, H\}, \tag{44}$$

$$1 = R_t^d \, E_t \left[\Lambda_{t,t+1} \Pi_{t+1}^{-1}\right], \tag{45}$$

where $\Pi_t \equiv P_t/P_{t-1}$ is gross inflation. Equation (42) defines marginal utility of consumption μ_t, Equation (43) is the stochastic discount factor, Equation (44) is the sector-specific labor supply condition, and Equation (45) is the Euler condition for the deposit composite (with R_t^d given by (40)).

A.2 Labor Market

The labor market is divided into two stages *within each sector* $s \in \{L, H\}$. In the first stage, there are many individual labor unions (h). These unions buy labor from households at the sectoral payment $MRS_{s,t}$ and then resell it to a central sectoral

"labor packer." The amount each union (h) buys, $L_{s,t}(h)$, is the same as the amount it sells, $L_{d,s,t}(h)$. In the second stage, the sectoral packer combines differentiated labor into sectoral aggregate labor input $L_{d,s,t}$ using CES technology with elasticity $\epsilon_w > 1$. The demand curve facing each union is

$$L_{d,s,t}(h) = \left(\frac{W_{s,t}(h)}{W_{s,t}}\right)^{-\epsilon_w} L_{d,s,t}, \tag{46}$$

and the sectoral wage index is

$$W_{s,t}^{1-\epsilon_w} = \int_0^1 W_{s,t}(h)^{1-\epsilon_w}\, dh. \tag{47}$$

The profit of a typical labor union in sector s (nominal terms) is

$$DIV_{L,s,t}(h) = W_{s,t}(h) L_{d,s,t}(h) - MRS_{s,t} L_{s,t}(h). \tag{48}$$

Wages are sticky (Calvo): in any period, a union can reset with probability $1-\phi_w$; otherwise it indexes to *sectoral* inflation at rate γ_w. Let $\Pi_{s,t}$ denote gross sectoral inflation. If a reset wage set at t is still in place at $t+j$, then (after indexation) it equals

$$W_{s,t+j}(h) = W_{s,t}(h) \left(\frac{P_{s,t+j-1}}{P_{s,t-1}}\right)^{\gamma_w}, \tag{49}$$

equivalently in terms of sectoral inflation.

Dropping h (reset wages are common across unions in each sector), the optimal reset *real* wage satisfies

$$w_{s,t}^* = \frac{\epsilon_w}{\epsilon_w - 1} \frac{f_{1,s,t}}{f_{2,s,t}}, \tag{50}$$

with recursions

$$f_{1,s,t} = mrs_{s,t}\, w_{s,t}^{\epsilon_w} L_{d,s,t} + \phi_w\, \mathbb{E}_t \left[\Lambda_{t,t+1} \left(\frac{\Pi_{s,t+1}}{\Pi_{s,t}^{\gamma_w}}\right)^{\epsilon_w} f_{1,s,t+1}\right], \tag{51}$$

$$f_{2,s,t} = w_{s,t}^{\epsilon_w} L_{d,s,t} + \phi_w\, \mathbb{E}_t \left[\Lambda_{t,t+1} \left(\frac{\Pi_{s,t+1}}{\Pi_{s,t}^{\gamma_w}}\right)^{\epsilon_w - 1} f_{2,s,t+1}\right], \tag{52}$$

for $s \in \{L, H\}$.

Aggregation. Integrating (46) across unions within each sector yields

$$L_{s,t} = L_{d,s,t} v^w_{s,t}, \tag{53}$$

where sectoral wage dispersion is

$$v^w_{s,t} = (1-\phi_w)\left(\frac{w^*_{s,t}}{w_{s,t}}\right)^{-\epsilon_w} + \phi_w \left(\frac{\Pi_{s,t}}{\Pi^{\gamma_w}_{s,t-1}}\right)^{\epsilon_w}\left(\frac{w_{s,t}}{w_{s,t-1}}\right)^{\epsilon_w} v^w_{s,t-1}. \tag{54}$$

Sectoral wage index evolution implies

$$w^{1-\epsilon_w}_{s,t} = (1-\phi_w)(w^*_{s,t})^{1-\epsilon_w} + \phi_w \Pi^{\gamma_w(1-\epsilon_w)}_{s,t-1}\Pi^{\epsilon_w-1}_{s,t} w^{1-\epsilon_w}_{s,t-1}. \tag{55}$$

Finally, total labor is the sum across sectors:

$$L_t = L_{L,t} + L_{H,t}. \tag{56}$$

A.3 Rest of Production

A.3.1 Capital Producer

Investment and installed capital are sector-specific. For $s \in \{L, H\}$, new effective capital in sector s is produced from final output subject to adjustment costs and an investment-efficiency shifter $\mu_{I,t}$:

$$\hat{I}_{s,t} \;=\; \mu_{I,t}\Big[1 - S_s\big(I_{s,t}/I_{s,t-1}\big)\Big] I_{s,t}, \qquad S_s(x) = \frac{\kappa_{I,s}}{2}(x-1)^2. \tag{57}$$

The optimality condition for the capital producer in segment s is

$$\begin{aligned} 1 \;&= p^k_{s,t}\mu_{I,t}\left[1 - \frac{\kappa_{I,s}}{2}\Big(\frac{I_{s,t}}{I_{s,t-1}} - 1\Big)^2 - \kappa_{I,s}\Big(\frac{I_{s,t}}{I_{s,t-1}} - 1\Big)\Big(\frac{I_{s,t}}{I_{s,t-1}}\Big)\right] \\ &\quad + E_t\left[\Lambda_{t,t+1}\, p^k_{s,t+1}\mu_{I,t+1}\,\kappa_{I,s}\Big(\frac{I_{s,t+1}}{I_{s,t}} - 1\Big)\Big(\frac{I_{s,t+1}}{I_{s,t}}\Big)^2\right], \end{aligned} \tag{58}$$

for $s \in \{L, H\}$. Aggregate investment is

$$I_t = I_{L,t} + I_{H,t}. \tag{59}$$

A.3.2 Retailers

Retail pricing is sector-specific. A retail firm f in sector s earns nominal profits

$$\mathrm{DIV}^R_{s,t}(f) = P_{s,t}(f)Y_{s,t}(f) - P_{m,s,t}Y_{s,t}(f), \tag{60}$$

and faces demand

$$Y_{s,t}(f) = \left(\frac{P_{s,t}(f)}{P_{s,t}}\right)^{-\epsilon_p} Y_{s,t}, \qquad \epsilon_p > 1. \tag{61}$$

With Calvo pricing, measure $1 - \phi_{p,s}$ of sector-s retailers can reset in t; otherwise, prices are indexed to lagged sectoral inflation at rate γ_p:

$$P_{s,t+j}(f) = P_{s,t}(f)\left(\frac{P_{s,t+j-1}}{P_{s,t-1}}\right)^{\gamma_p}. \tag{62}$$

A sector-s retailer chooses $P_{s,t}(f)$ to maximize the present discounted value of real profits returned to households:

$$\begin{aligned}\max_{P_{s,t}(f)} E_t \sum_{j\geq 0} \phi^j_{p,s}\Lambda_{t,t+j}\Bigg[& P_{s,t}(f)^{1-\epsilon_p}\left(\frac{P_{s,t+j-1}}{P_{s,t-1}}\right)^{(1-\epsilon_p)\gamma_p} P^{\epsilon_p-1}_{s,t+j}Y_{s,t+j} \\ & - P_{m,s,t+j}P_{s,t}(f)^{-\epsilon_p}\left(\frac{P_{s,t+j-1}}{P_{s,t-1}}\right)^{-\epsilon_p\gamma_p} P^{\epsilon_p}_{s,t+j}Y_{s,t+j}\Bigg].\end{aligned} \tag{63}$$

The FOC implies a sectoral reset price

$$P^*_{s,t} = \frac{\epsilon_p}{\epsilon_p - 1}\frac{X_{1,s,t}}{X_{2,s,t}}, \tag{64}$$

or in relative terms $p^*_{s,t} \equiv P^*_{s,t}/P_{s,t}$,

$$p^*_{s,t} = \frac{\epsilon_p}{\epsilon_p - 1}\frac{x_{1,s,t}}{x_{2,s,t}}, \tag{65}$$

with $x_{1,s,t} \equiv X_{1,s,t}/P^{\epsilon_p}_{s,t}$ and $x_{2,s,t} \equiv X_{2,s,t}/P^{\epsilon_p-1}_{s,t}$ satisfying

$$x_{1,s,t} = p_{m,s,t}\,Y_{s,t} + \phi_{p,s}E_t\left[\Lambda_{t,t+1}\left(\frac{\Pi_{s,t+1}}{\Pi^{\gamma_p}_{s,t}}\right)^{\epsilon_p} x_{1,s,t+1}\right], \tag{66}$$

$$x_{2,s,t} = Y_{s,t} + \phi_{p,s} E_t \left[\Lambda_{t,t+1} \left(\frac{\Pi_{s,t+1}}{\Pi_{s,t}^{\gamma_p}} \right)^{\epsilon_p - 1} x_{2,s,t+1} \right], \tag{67}$$

for $s \in \{L, H\}$, where ζ_s are the sectoral weights from the final-goods aggregator.

Aggregation yields sectoral price dispersion $v_{s,t}^p$ and the wholesale–final output link:

$$Y_{m,s,t} = Y_{s,t} v_{s,t}^p, \tag{68}$$

$$v_{s,t}^p = (1 - \phi_{p,s})(p_{s,t}^*)^{-\epsilon_p} + \phi_{p,s} \left(\frac{\Pi_{s,t}}{\Pi_{s,t-1}^{\gamma_p}} \right)^{\epsilon_p} v_{s,t-1}^p, \tag{69}$$

and the sectoral price index evolves as

$$1 = (1 - \phi_{p,s})(p_{s,t}^*)^{1-\epsilon_p} + \phi_{p,s} \, \Pi_{s,t-1}^{\gamma_p(1-\epsilon_p)} \Pi_{s,t}^{\epsilon_p - 1}, \tag{70}$$

for $s \in \{L, H\}$.

A.4 Fiscal Authority

The fiscal authority purchases an exogenous stochastic quantity of final goods G_t in each period. It finances this with lump-sum taxes T_t, remittances from the central bank $T_{cb,t}$, and issuing long-term nominal government bonds. Let $Q_{B,t}$ be the price of the long-term government bond (perpetuity with coupon decay $\kappa_t^b \in [0, 1]$), and let $b_{G,t}$ denote the real quantity of government bonds outstanding at the end of period t (so the beginning-of-period stock is $b_{G,t-1}$). The government's real budget constraint is

$$G_t + \Pi_t^{-1} b_{G,t-1} = T_t + T_{cb,t} + Q_{B,t} \left(b_{G,t} - \kappa_t^b \, \Pi_t^{-1} b_{G,t-1} \right). \tag{71}$$

The transfer $T_{cb,t}$ is determined by the central bank balance sheet and remittance rule in the monetary authority block.

A.5 Aggregation and Exogenous Processes

The model features sector-specific and aggregate exogenous processes. For $s \in \{L, H\}$, productivity and financial tightness follow AR(1) processes in logs:

$$\log A_{s,t} = \rho_{A,s} \log A_{s,t-1} + s_{A,s}\varepsilon_{A,s,t}, \tag{72}$$

$$\log \theta_{s,t} = (1-\rho_\theta)\log \bar{\theta}_s + \rho_\theta \ln \theta_{s,t-1} + s_{\theta,s}\varepsilon_{\theta,s,t}, \tag{73}$$

and sectoral haircuts (or relative pledgeability) follow

$$\log \Delta_{s,t} = (1-\rho_D)\log \bar{\Delta}_s + \rho_D \log \Delta_{s,t-1} + s_{\Delta,s}\varepsilon_{\Delta,s,t}. \tag{74}$$

Aggregate government spending and debt supply follow

$$\log G_t = (1-\rho_G)\log \bar{G} + \rho_G \log G_{t-1} + s_G\varepsilon_{G,t}, \tag{75}$$

$$\log b_{G,t} = (1-\rho_B)\log \bar{b}_G + \rho_B \ln b_{G,t-1} + s_B\varepsilon_{B,t}. \tag{76}$$

Preference and investment-efficiency shifters follow

$$\log \mu_{I,t} = \rho_m \log \mu_{I,t-1} + s_m\varepsilon_{m,t}, \tag{77}$$

$$\log \psi_t^c = \rho_c \log \psi_{t-1}^c + s_c\varepsilon_{c,t}, \tag{78}$$

$$\log \psi_t^l = \rho_l \log \psi_{t-1}^l + s_l\varepsilon_{l,t}. \tag{79}$$

Sectoral maturities follow AR(1) processes in logs. For $s \in \{L, H\}$:

$$\log M_{s,t}^f = (1-\rho_{kf})\log \bar{M}_s^f + \rho_{kf} \log M_{s,t-1}^f + s_{kf,s}\varepsilon_{kf,s,t}, \tag{80}$$

$$\log M_t^b = (1-\rho_{kb})\log \bar{M}^b + \rho_{kb} \log M_{t-1}^b + s_{kb}\varepsilon_{kb,t}. \tag{81}$$

The decay parameters governing the coupon payments of the long-term bonds are derived as:

$$\kappa_{s,t}^f = 1 - \frac{1}{M_{s,t}^f} \quad \text{and} \quad \kappa_t^b = 1 - \frac{1}{M_t^b}. \tag{82}$$

Final output is a CES composite across sectors with weights (ζ_L, ζ_H) and substitution

elasticity η_y:

$$Y_t = \left[\zeta_L^{\frac{1}{\eta_y}} Y_{L,t}^{\frac{\eta_y - 1}{\eta_y}} + \zeta_H^{\frac{1}{\eta_y}} Y_{H,t}^{\frac{\eta_y - 1}{\eta_y}} \right]^{\frac{\eta_y}{\eta_y - 1}}. \tag{83}$$

The aggregate price index is normalized to 1 ($P_t = 1$), implying the following relation between relative sectoral prices $p_{s,t} \equiv P_{s,t}/P_t$:

$$1 = \zeta_L \, p_{L,t}^{1-\eta_y} + \zeta_H \, p_{H,t}^{1-\eta_y}. \tag{84}$$

Sectoral demand (expenditure share) satisfies:

$$Y_{s,t} = \zeta_s Y_t \, p_{s,t}^{-\eta_y}, \qquad s \in \{H, L\}. \tag{85}$$

Bond market clearing is sectoral for private bonds and aggregate for government bonds:

$$f_{s,t}^m = f_{s,t} + f_{s,t}^{cb}, \qquad s \in \{L, H\}, \tag{86}$$

$$b_{G,t} = b_{L,t} + b_{H,t} + b_t^{cb}, \tag{87}$$

and aggregate positions are $f_t = f_{L,t} + f_{H,t}$, $b_t = b_{L,t} + b_{H,t}$.

Intermediary balance sheets (real terms) hold separately for each sector/intermediary type:

$$Q_{s,t} f_{s,t} + Q_{B,t} b_{s,t} + re_{s,t} = d_{s,t} + n_{s,t}, \qquad s \in \{L, H\}. \tag{88}$$

Net worth evolves as

$$\begin{aligned} n_{s,t} = \sigma_s \Pi_t^{-1} \Big[& (R_{s,t}^F - R_{s,t-1}^d) Q_{s,t-1} f_{s,t-1} + (R_t^B - R_{s,t-1}^d) Q_{B,t-1} b_{s,t-1} \\ & + (R_{t-1}^{re} - R_{s,t-1}^d) re_{s,t-1} + R_{s,t-1}^d n_{s,t-1} \Big] + X_s, \qquad s \in \{L, H\}. \end{aligned} \tag{89}$$

The (binding) leverage constraint is

$$Q_{s,t} f_{s,t} + \Delta_{s,t} Q_{B,t} b_{s,t} = \phi_{s,t} n_{s,t}, \qquad s \in \{L, H\}. \tag{90}$$

Aggregate reserves satisfies

$$re_t = re_{L,t} + re_{H,t} \tag{91}$$

Aggregate net worth satisfies

$$n_t = n_{L,t} + n_{H,t} \tag{92}$$

Aggregate deposits satisfies

$$d_t = d_{L,t} + d_{H,t} \tag{93}$$

Resource constraint:

$$Y_t = C_t + I_t + G_t, \tag{94}$$

with $I_t = I_{L,t} + I_{H,t}$.

Appendix B Equilibrium Conditions

Notation. The economy has two segments, Low vs. High maturity gap, indexed by $s \in \{L, H\}$. Expectations are $\mathbf{E}_t[\cdot]$.

Households

$$\mu_t = \frac{\psi_{c,t}}{C_t - b\,C_{t-1}} - \beta b\,\frac{\psi_{c,t+1}}{C_{t+1} - b\,C_t}. \tag{B.1}$$

$$\psi_{c,t}\psi_{l,t}\,\chi_s\,L_{s,t}^{\eta} = \mu_t\,mrs_{s,t}, \qquad s \in \{L, H\}. \tag{B.2}$$

$$R_t^d\,\mathbf{E}_t\big[\Lambda_{t,t+1}\,\Pi_{t+1}^{-1}\big] = 1. \tag{B.3}$$

$$\frac{d_{L,t}}{d_{H,t}} = \left(\frac{\omega_d}{1-\omega_d}\right)^{\eta_d}\left(\frac{R_{L,t}^d}{R_{H,t}^d}\right)^{\eta_d}. \tag{B.4}$$

$$\left(R_t^d\right)^{1-\eta_d} = \omega_d\left(R_{L,t}^d\right)^{1-\eta_d} + (1-\omega_d)\left(R_{H,t}^d\right)^{1-\eta_d}. \tag{B.5}$$

$$\Lambda_{t,t+1} = \beta\,\frac{\mu_{t+1}}{\mu_t}. \tag{B.6}$$

Financial Intermediaries

$$R_{s,t}^F = \frac{1+\kappa_{s,t}^f\,Q_{s,t}}{Q_{s,t-1}}, \qquad s \in \{L, H\}. \tag{B.7}$$

$$R_t^B = \frac{1+\kappa_t^b\,Q_{B,t}}{Q_{B,t-1}}. \tag{B.8}$$

$$\Omega_{s,t} = 1 - \sigma_s + \sigma_s\,\theta_{s,t}\,\phi_{s,t}, \qquad s \in \{L, H\}. \tag{B.9}$$

$$\mathbf{E}_t\big[\Lambda_{t,t+1}\,\Omega_{s,t+1}\,\Pi_{t+1}^{-1}\left(R_{s,t+1}^F - R_{s,t}^d\right)\big] = \frac{\lambda_{s,t}}{1+\lambda_{s,t}}\,\theta_{s,t}, \qquad s \in \{L, H\}. \tag{B.10}$$

$$\mathbf{E}_t\left[\Lambda_{t,t+1}\,\Omega_{s,t+1}\,\Pi_{t+1}^{-1}\left(R_{t+1}^{B}-R_{s,t}^{d}\right)\right]=\frac{\lambda_{s,t}}{1+\lambda_{s,t}}\,\theta_{s,t}\,\Delta_{s,t},\qquad s\in\{L,H\}. \tag{B.11}$$

$$\mathbf{E}_t\left[\Lambda_{t,t+1}\,\Omega_{s,t+1}\,\Pi_{t+1}^{-1}\left(R_{t}^{re}-R_{s,t}^{d}\right)\right]=0,\qquad s\in\{L,H\}. \tag{B.12}$$

$$\theta_{s,t}\,\phi_{s,t}=(1+\lambda_{s,t})\,\mathbf{E}_t\left[\Lambda_{t,t+1}\,\Omega_{s,t+1}\,\Pi_{t+1}^{-1}\,R_{s,t}^{d}\right],\qquad s\in\{L,H\}. \tag{B.13}$$

Labor Market

Labor Unions

$$w_{s,t}^{*}=\frac{\epsilon_w}{\epsilon_w-1}\frac{f_{1,s,t}}{f_{2,s,t}},\qquad s\in\{L,H\}. \tag{B.14}$$

$$f_{1,s,t}=mrs_{s,t}\,w_{s,t}^{\epsilon_w}\,L_{s,t}^{d}+\phi_w\,\mathbf{E}_t\left[\Lambda_{t,t+1}\,\Pi_{s,t+1}^{\epsilon_w}\,\Pi_{s,t}^{-\epsilon_w\gamma_w}\,f_{1,s,t+1}\right],\qquad s\in\{L,H\}. \tag{B.15}$$

$$f_{2,s,t}=w_{s,t}^{\epsilon_w}\,L_{s,t}^{d}+\phi_w\,\mathbf{E}_t\left[\Lambda_{t,t+1}\,\Pi_{s,t+1}^{\epsilon_w-1}\,\Pi_{s,t}^{(1-\epsilon_w)\gamma_w}\,f_{2,s,t+1}\right],\qquad s\in\{L,H\}. \tag{B.16}$$

Aggregation

$$L_{s,t}=L_{s,t}^{d}\,v_{s,t}^{w},\qquad s\in\{L,H\}. \tag{B.17}$$

$$L_t=L_{L,t}+L_{H,t}. \tag{B.18}$$

$$v_{s,t}^{w}=(1-\phi_w)\left(\frac{w_{s,t}^{*}}{w_{s,t}}\right)^{-\epsilon_w}+\phi_w\,\Pi_{s,t}^{\epsilon_w}\,\Pi_{s,t-1}^{-\gamma_w\epsilon_w}\,w_{s,t}^{\epsilon_w}\,w_{s,t-1}^{-\epsilon_w}\,v_{s,t-1}^{w},\qquad s\in\{L,H\}. \tag{B.19}$$

$$w_{s,t}^{1-\epsilon_w}=(1-\phi_w)\,(w_{s,t}^{*})^{1-\epsilon_w}+\phi_w\,\Pi_{s,t-1}^{\gamma_w(1-\epsilon_w)}\,\Pi_{s,t}^{\epsilon_w-1}\,w_{s,t-1}^{1-\epsilon_w},\qquad s\in\{L,H\}. \tag{B.20}$$

Production

Retail Firms

$$p^*_{s,t} = \frac{\epsilon_p}{\epsilon_p - 1} \frac{x_{1,s,t}}{x_{2,s,t}}, \qquad s \in \{L, H\}. \tag{B.21}$$

$$x_{1,s,t} = p^m_{s,t}\, Y_{s,t} + \phi_{p,s}\, \mathbf{E}_t \Big[\Lambda_{t,t+1}\, \Pi^{\epsilon_p}_{s,t+1}\, \Pi^{-\epsilon_p \gamma_p}_{s,t}\, x_{1,s,t+1} \Big], \qquad s \in \{L, H\}. \tag{B.22}$$

$$x_{2,s,t} = Y_{s,t} + \phi_{p,s}\, \mathbf{E}_t \Big[\Lambda_{t,t+1}\, \Pi^{\epsilon_p - 1}_{s,t+1}\, \Pi^{(1-\epsilon_p)\gamma_p}_{s,t}\, x_{2,s,t+1} \Big], \qquad s \in \{L, H\}. \tag{B.23}$$

Wholesale Firms

$$Y^m_{s,t} = A_{s,t}\, (u_{s,t} K_{s,t-1})^{\alpha}\, (L^d_{s,t})^{1-\alpha}, \qquad s \in \{L, H\}. \tag{B.24}$$

$$w_{s,t} = p^m_{s,t} (1-\alpha)\, A_{s,t}\, (u_{s,t} K_{s,t-1})^{\alpha}\, (L^d_{s,t})^{-\alpha}, \qquad s \in \{L, H\}. \tag{B.25}$$

$$p^k_{s,t} M_{1,s,t} \Big(\delta_{1,s} + \delta_2 (u_{s,t} - 1) \Big) = \alpha\, p^m_{s,t}\, A_{s,t}\, (u_{s,t} K_{s,t-1})^{\alpha - 1}\, (L^d_{s,t})^{1-\alpha}, \qquad s \in \{L, H\}. \tag{B.26}$$

$$\begin{aligned} p^k_{s,t} M_{1,s,t} = \mathbf{E}_t \Bigg[\Lambda_{t,t+1} \Big(& \alpha\, p^m_{s,t+1} A_{s,t+1}\, K^{\alpha-1}_{s,t}\, u^{\alpha}_{s,t+1}\, (L^d_{s,t+1})^{1-\alpha} \\ & + \Big(1 - \delta_{0,s} - \delta_{1,s}(u_{s,t+1} - 1) - \frac{\delta_2}{2} (u_{s,t+1} - 1)^2 \Big)\, p^k_{s,t+1} M_{1,s,t+1} \Big) \Bigg], \end{aligned} \qquad s \in \{L, H\}. \tag{B.27}$$

$$Q_{s,t} M_{2,s,t} = \mathbf{E}_t \Big[\Lambda_{t,t+1}\, \Pi^{-1}_{s,t+1} \Big(1 + \kappa^f_{s,t}\, Q_{s,t+1} M_{2,s,t+1} \Big) \Big], \qquad s \in \{L, H\}. \tag{B.28}$$

$$\frac{M_{1,s,t} - 1}{M_{2,s,t} - 1} = \psi_s, \qquad s \in \{L, H\}. \tag{B.29}$$

$$\psi_s \, p^k_{s,t} \, \hat{I}_{s,t} = Q_{s,t}\Big(f^m_{s,t} - \kappa^f_{s,t} \, f^m_{s,t-1} \, \Pi^{-1}_{s,t} \Big), \qquad s \in \{L, H\}. \tag{B.30}$$

$$K_{s,t} = \hat{I}_{s,t} + \Big(1 - \delta_{0,s} - \delta_{1,s}(u_{s,t} - 1) - \frac{\delta_2}{2}(u_{s,t} - 1)^2\Big) K_{s,t-1}, \qquad s \in \{L, H\}. \tag{B.31}$$

$$K_t = K_{L,t} + K_{H,t}. \tag{B.32}$$

Capital Producers

$$\hat{I}_{s,t} = \mu_{I,t} \left(1 - \frac{\kappa_{I,s}}{2} \left(\frac{I_{s,t}}{I_{s,t-1}} - 1\right)^2\right) I_{s,t}, \qquad s \in \{L, H\}. \tag{B.33}$$

$$\begin{aligned} 1 = {} & p^k_{s,t} \mu_{I,t} \left(1 - \frac{\kappa_{I,s}}{2} \left(\frac{I_{s,t}}{I_{s,t-1}} - 1\right)^2 - \kappa_{I,s} \left(\frac{I_{s,t}}{I_{s,t-1}} - 1\right) \left(\frac{I_{s,t}}{I_{s,t-1}}\right)\right) \\ & + \mathbf{E}_t \left[\Lambda_{t,t+1} \, p^k_{s,t+1} \mu_{I,t+1} \, \kappa_{I,s} \left(\frac{I_{s,t+1}}{I_{s,t}} - 1\right) \left(\frac{I_{s,t+1}}{I_{s,t}}\right)^2\right], \qquad s \in \{L, H\}. \end{aligned} \tag{B.34}$$

$$I_t = I_{L,t} + I_{H,t}. \tag{B.35}$$

Government

$$G_t + \Pi_t^{-1} b_{G,t-1} = T_t + T_{cb,t} + Q_{B,t}\Big(b_{G,t} - \kappa^b_t \, \Pi_t^{-1} b_{G,t-1} \Big). \tag{B.36}$$

Central Bank and Monetary Policy

$$Q_{L,t} f_{cb,L,t} + Q_{H,t} f_{cb,H,t} + Q_{B,t} b_{cb,t} = re_t. \tag{B.37}$$

$$\begin{aligned} T_{cb,t} = {} & (1 + \kappa^f_{L,t} Q_{L,t}) \, \Pi_t^{-1} f_{cb,L,t-1} + (1 + \kappa^f_{H,t} Q_{H,t}) \, \Pi_t^{-1} f_{cb,H,t-1} \\ & + (1 + \kappa^b_t Q_{B,t}) \, \Pi_t^{-1} b_{cb,t-1} - R^{re}_{t-1} \, \Pi_t^{-1} re_{t-1}. \end{aligned} \tag{B.38}$$

$$\begin{aligned}\log R_t^{tr} &= (1-\rho_r)\log R^{tr} + \rho_r \log R_{t-1}^{tr} \\ &\quad + (1-\rho_r)\Big[\phi_\pi(\log \Pi_t - \log \Pi) + \phi_y(\log Y_t - \log Y_{t-1})\Big] + s_r \varepsilon_{r,t}.\end{aligned} \tag{B.39}$$

$$R_t^{re} = R_t^{tr}. \tag{B.40}$$

Aggregation, Price System, and Market Clearing

$$Y_{s,t} = \zeta_s \, Y_t \, p_{s,t}^{-\eta_y}, \qquad s \in \{L, H\}. \tag{B.41}$$

$$Y_{s,t}^m = Y_{s,t} \, v_{s,t}^p, \qquad s \in \{L, H\}. \tag{B.42}$$

$$v_{s,t}^p = (1-\phi_{p,s})(p_{s,t}^*)^{-\epsilon_p} + \phi_{p,s} \, \Pi_{s,t}^{\epsilon_p} \, \Pi_{s,t-1}^{-\gamma_p \epsilon_p} \, v_{s,t-1}^p, \qquad s \in \{L, H\}. \tag{B.43}$$

$$1 = (1-\phi_{p,s})(p_{s,t}^*)^{1-\epsilon_p} + \phi_{p,s} \, \Pi_{s,t}^{\epsilon_p - 1} \, \Pi_{s,t-1}^{\gamma_p(1-\epsilon_p)}, \qquad s \in \{L, H\}. \tag{B.44}$$

$$\Pi_{s,t} = \Pi_t \left(\frac{p_{s,t}}{p_{s,t-1}} \right), \qquad s \in \{L, H\}. \tag{B.45}$$

$$Y_t = \left[\zeta_L^{\frac{1}{\eta_y}} Y_{L,t}^{\frac{\eta_y - 1}{\eta_y}} + \zeta_H^{\frac{1}{\eta_y}} Y_{H,t}^{\frac{\eta_y - 1}{\eta_y}} \right]^{\frac{\eta_y}{\eta_y - 1}}. \tag{B.46}$$

$$Q_{s,t} f_{s,t} + Q_{B,t} b_{s,t} + re_{s,t} = d_{s,t} + n_{s,t}, \qquad s \in \{L, H\}. \tag{B.47}$$

$$Q_{s,t} f_{s,t} + \Delta_{s,t} Q_{B,t} b_{s,t} = \phi_{s,t} n_{s,t}, \qquad s \in \{L, H\}. \tag{B.48}$$

$$\begin{aligned} n_{s,t} = \sigma_s \, \Pi_t^{-1} \Big(&(R_{s,t}^F - R_{s,t-1}^d) \, Q_{s,t-1} f_{s,t-1} + (R_t^B - R_{s,t-1}^d) \, Q_{B,t-1} b_{s,t-1} \\ &+ (R_{t-1}^{re} - R_{s,t-1}^d) \, re_{s,t-1} + R_{s,t-1}^d \, n_{s,t-1} \Big) + X_s, \qquad s \in \{L, H\}. \end{aligned} \tag{B.49}$$

$$b_{G,t} = b_t + b_{cb,t}. \tag{B.50}$$

$$b_t = b_{L,t} + b_{H,t}. \tag{B.51}$$

$$f^m_{s,t} = f_{s,t} + f_{cb,s,t}, \qquad s \in \{L, H\}. \tag{B.52}$$

$$f_t = f_{L,t} + f_{H,t}. \tag{B.53}$$

$$d_t = d_{L,t} + d_{H,t}. \tag{B.54}$$

$$n_t = n_{L,t} + n_{H,t}. \tag{B.55}$$

$$Y_t = C_t + I_t + G_t. \tag{B.56}$$

$$re_t = re_{L,t} + re_{H,t}. \tag{B.57}$$

$$R^{L,F}_{s,t} = \kappa^f_{s,t} + Q^{-1}_{s,t}, \qquad s \in \{L, H\}. \tag{B.58}$$

$$R^{L,B}_t = \kappa^b_t + Q^{-1}_{B,t}. \tag{B.59}$$

Exogenous Processes (Shocks)

$$\log A_{s,t} = \rho_{A,s} \log A_{s,t-1} + s_{A,s}\, \varepsilon_{A,s,t}, \qquad s \in \{L, H\}. \tag{B.60}$$

$$\log \mu_{I,t} = \rho_m \log \mu_{I,t-1} + s_m\, \varepsilon_{m,t}. \tag{B.61}$$

$$\log \psi_{c,t} = \rho_c \log \psi_{c,t-1} + s_c\, \varepsilon_{c,t}. \tag{B.62}$$

$$\log \psi_{l,t} = \rho_l \log \psi_{l,t-1} + s_l\, \varepsilon_{l,t}. \tag{B.63}$$

$$\log G_t = (1-\rho_G)\log G + \rho_G \log G_{t-1} + s_G\,\varepsilon_{G,t}. \tag{B.64}$$

$$\log b_{G,t} = (1-\rho_B)\log \bar{b}_G + \rho_B \log b_{G,t-1} + s_B\,\varepsilon_{B,t}. \tag{B.65}$$

$$b_{cb,t} = (1-\rho_b)b_{cb} + \rho_b b_{cb,t-1} + s_b\,\varepsilon_{b,t}. \tag{B.66}$$

$$f_{cb,s,t} = (1-\rho_f)f_{cb,s} + \rho_f f_{cb,s,t-1} + s_{f,s}\,\varepsilon_{f,s,t}, \qquad s \in \{L,H\}. \tag{B.67}$$

$$\log \theta_{s,t} = (1-\rho_\theta)\log \bar{\theta}_s + \rho_\theta \log \theta_{s,t-1} + s_{\theta,s}\,\varepsilon_{\theta,s,t}, \qquad s \in \{L,H\}. \tag{B.68}$$

$$\log \Delta_{s,t} = (1-\rho_D)\log \bar{\Delta}_s + \rho_D \log \Delta_{s,t-1} + s_{D,s}\,\varepsilon_{D,s,t}, \qquad s \in \{L,H\}. \tag{B.69}$$

Returns and Spreads

$$return^f_{s,t} = R^F_{s,t} - R^d_{s,t-1}, \qquad s \in \{L,H\}. \tag{B.70}$$

$$return^b_{s,t} = R^B_t - R^d_{s,t-1}, \qquad s \in \{L,H\}. \tag{B.71}$$

$$Spread^f_{s,t} = R^{L,F}_{s,t} - R^d_{s,t}, \qquad s \in \{L,H\}. \tag{B.72}$$

$$Spread^b_{s,t} = R^{L,B}_t - R^d_{s,t}, \qquad s \in \{L,H\}. \tag{B.73}$$

$$return^{fb}_{s,t} = R^F_{s,t} - R^B_t, \qquad s \in \{L,H\}. \tag{B.74}$$

Real Rates

$$f^r_{s,t} = R^{L,F}_{s,t} - \Pi_{t+1}, \qquad s \in \{L,H\}. \tag{B.75}$$

$$b_t^r = R_t^{L,B} - \Pi_{t+1}. \tag{B.76}$$

Welfare

$$W_t = \log(C_t - b\,C_{t-1}) - \chi_L \frac{L_{L,t}^{1+\eta}}{1+\eta} - \chi_H \frac{L_{H,t}^{1+\eta}}{1+\eta} + \beta\,\mathbf{E}_t[W_{t+1}]. \tag{B.77}$$

Maturity Gap and Market Values

$$\log M_{s,t}^f = (1-\rho_{kf})\log \bar{M}_s^f + \rho_{kf}\log M_{s,t-1}^f + s_{kf,s}\,\varepsilon_{kf,s,t}, \qquad s \in \{L,H\}. \tag{B.78}$$

$$\log M_t^b = (1-\rho_{kb})\log \bar{M}^b + \rho_{kb}\log M_{t-1}^b + s_{kb}\,\varepsilon_{kb,t}. \tag{B.79}$$

$$\kappa_{s,t}^f = 1 - \frac{1}{M_{s,t}^f}, \qquad s \in \{L,H\}. \tag{B.80}$$

$$\kappa_t^b = 1 - \frac{1}{M_t^b}. \tag{B.81}$$

$$mgap_{s,t} = \left(\frac{M_{s,t}^f\,f_{s,t}\,Q_{s,t} + M_t^b\,b_{s,t}\,Q_{B,t} + M_s^{re}\,re_{s,t}}{f_{s,t}\,Q_{s,t} + b_{s,t}\,Q_{B,t} + re_{s,t}}\right) - M_s^d, \qquad s \in \{L,H\}. \tag{B.82}$$

$$MV_{s,t}^f = Q_{s,t}\,f_{s,t}, \qquad s \in \{L,H\}. \tag{B.83}$$

$$MV_t^f = MV_{L,t}^f + MV_{H,t}^f. \tag{B.84}$$

$$MV_{s,t}^b = Q_{B,t}\,b_{s,t}, \qquad s \in \{L,H\}. \tag{B.85}$$

$$MV_t^b = MV_{L,t}^b + MV_{H,t}^b. \tag{B.86}$$

The full set of equilibrium conditions consists of **137 equations for 137 endogenous variables**. All equations indexed by $s \in \{L,H\}$ represent **two** segments (Low and

High).

- **Households**: marginal utility, labor FOCs (s), deposit Euler + aggregation, SDF.
- **Financial Intermediaries**: returns, $\Omega_{s,t}$, portfolio FOCs (s), multiplier link (s).
- **Labor Market**: wage setting (s), wage dispersion and wage evolution (s), employment aggregation.
- **Production**: retail pricing (s), wholesale block (s), capital producer block (s).
- **Government and Central Bank**: fiscal constraint, CB balance sheet, remittances, Taylor rule, rate setting.
- **Aggregation and Market Clearing**: output/price system, intermediary constraints, bond clearing, resource constraint, reserves, long yields.
- **Exogenous Processes**: sectoral and aggregate AR(1) shocks.
- **Definitions**: spreads/returns, real rates, reporting variables, maturity gap and market values.
- **Welfare**: recursive welfare.

These comprise 137 equations for the 137 endogenous variables:

$$\{C_t, L_t, L_{L,t}, L_{H,t}, mrs_{L,t}, mrs_{H,t}, R^d_{L,t}, R^d_{H,t}, R^d_t, \Lambda_{t,t+1}, \Pi_t, \Pi_{L,t}, \Pi_{H,t}, Y_t, G_t,$$
$$Q_{L,t}, Q_{H,t}, Q_{B,t}, f_{L,t}, f_{H,t}, f_t, b_{L,t}, b_{H,t}, b_t, d_t, d_{L,t}, d_{H,t}, n_t, n_{L,t}, n_{H,t}, R^F_{L,t}, R^F_{H,t}, R^B_t, R^{re}_t, R^{tr}_t,$$
$$\phi_{L,t}, \phi_{H,t}, f^m_{L,t}, f^m_{H,t}, MV^f_t, MV^f_{L,t}, MV^f_{H,t}, MV^b_t, MV^b_{L,t}, MV^b_{H,t}, w^*_{L,t}, w^*_{H,t}, w_{L,t}, w_{H,t}, L^d_{L,t}, L^d_{H,t},$$
$$p^*_{L,t}, p^*_{H,t}, Y^m_{L,t}, Y^m_{H,t}, A_{L,t}, A_{H,t}, K_t, K_{L,t}, K_{H,t}, u_{L,t}, u_{H,t}, p^k_{L,t}, p^k_{H,t}, M_{1,L,t}, M_{1,H,t}, M_{2,L,t}, M_{2,H,t},$$
$$\hat{I}_{L,t}, \hat{I}_{H,t}, I_{L,t}, I_{H,t}, I_t, \Delta_{L,t}, \Delta_{H,t}, \theta_{L,t}, \theta_{H,t}, \psi_{l,t}, \psi_{c,t}, \mu_{I,t}, R^{L,B}_t, R^{L,F}_{L,t}, R^{L,F}_{H,t}, b_{G,t}, re_{L,t}, re_{H,t}, re_t,$$
$$\mu_t, \Omega_{L,t}, \Omega_{H,t}, p^m_{L,t}, p^m_{H,t}, \lambda_{L,t}, \lambda_{H,t}, f_{1,L,t}, f_{1,H,t}, f_{2,L,t}, f_{2,H,t}, v^w_{L,t}, v^w_{H,t}, x_{1,L,t}, x_{1,H,t}, x_{2,L,t}, x_{2,H,t},$$
$$v^p_{L,t}, v^p_{H,t}, \kappa^f_{L,t}, \kappa^f_{H,t}, \kappa^b_t, M^f_{L,t}, M^f_{H,t}, M^b_t, mgap_{L,t}, mgap_{H,t}, T_t, T_{cb,t}, W_t, b_{cb,t}, f_{cb,L,t}, f_{cb,H,t}, f^r_{L,t}, f^r_{H,t},$$
$$b^r_t, return^f_{L,t}, return^f_{H,t}, return^b_{L,t}, return^b_{H,t}, return^{fb}_{L,t}, return^{fb}_{H,t}, Spread^b_{L,t}, Spread^b_{H,t},$$
$$Spread^f_{L,t}, Spread^f_{H,t}, Y_{L,t}, Y_{H,t}, p_{L,t}, p_{H,t}\}$$

Table 5: Calibrated Parameters: Standard and Aggregate

Param.	Value/Target	Description
Household & Technology		
β	0.995	Discount factor
b	0.70	Habit formation
η	1	Inverse Frisch elasticity
α	0.33	Capital share in production
δ_0	0.025	Steady-state depreciation rate
δ_2	0.01	Utilization adjustment cost
ϵ_p, ϵ_w	11	Elasticity of substitution (goods & labor)
ϕ_w	0.75	Wage rigidity (Calvo)
ϕ_p	0.75	Price Stickiness (Calvo)
σ	0.95	Intermediary survival prob.
ψ	0.80	Loan-in-advance constraint
κ_I	2.00	Investment adjustment cost
χ	1.00	Labor disutility scaling
γ_p, γ_w	0	Price and wage indexation
Π	1	Steady-state (gross) inflation
η_d	5	Elasticity of substitution (between deposits)
ω_d	0.65	Deposit preference weight
η_y	1.5	CES elasticity of substitution (between segments)
Fiscal & Central Bank		
G/Y	0.22	Government spending share
B/Y	3.20	Steady-state debt-to-GDP (4×0.80)
B_{cb}/Y	0.24	CB government bond holdings (4×0.06)
M^b	40	Target maturity: government bonds (quarters)
κ^b	$1 - 1/40$	Decay rate: government bonds
Monetary Policy		
ρ_r	0.80	Interest rate smoothing
ϕ_π	1.80	Response to inflation
ϕ_y	0.25	Response to output growth
s_r	0.10	Monetary policy shock volatility
Aggregate Shock Processes		
ρ_G, s_G	0.95, 0.01	Government spending
ρ_B, s_B	0.95, 0.00	Government debt supply
ρ_c, s_c	0.90, 0.01	Consumption preference shock
ρ_l, s_l	0.90, 0.00	Labor supply shock
ρ_m, s_m	0.90, 0.00	Investment efficiency shock
ρ_b, s_b	0.90, 0.10	CB government bond holdings
$\rho_{\kappa^b}, s_{\kappa^b}$	0.95, 0.01	Gov. bond duration shock

Note: Shock volatilities set to zero indicate that the corresponding shock is inactive in the baseline calibration. Price stickiness ($\phi_{p,s}$) and intermediary survival probability (σ_s) are segment-specific and reported in Table 4.

Appendix C Empirical analysis

Table 6: Description of the variables

Variable	Unit and Frequency	Data Source	Description
Dependent and core variables			
Loan growth	Percentages, monthly	AnaCredit	Computed from the (recognized) outstanding loans reported in *AnaCredit*, where the counterparty is a euro area non-financial corporation. Aggregations are done at the bank–month–economic activity of the counterparty (NACE sector) level.
Maturity gap	Years, quarterly	ECB Supervisory Reporting data	The maturity gap proxy is calculated based on the future cash flows, both inflows and outflows, reported by euro area banks in template COREP C66.01 – *Maturity Ladder*. Cash flows are reported in 21 maturity buckets. The maturity of inflows and outflows in each bucket is proxied by the bucket's midpoint (for example, 1.5 years for flows in "Greater than 12 months and up to 2 years"). A maturity of 15 years is assigned to cash flows allocated to "Greater than 5 years"[1]. The maturity-weighted difference between inflows and outflows is scaled by the bank's total assets (from the FINREP template F01).
Monetary policy shocks: Target and QT	Basis points, monthly	EA-MPD (Altavilla et al. (2019))	Conventional and unconventional monetary policy shocks refer to the "Target" and "QT" shocks, respectively, as defined by Altavilla et al. (2019). These are computed as the rotated factors explaining high-frequency changes in Overnight Index Swap (OIS) rates around monetary policy events (i.e., between the press release of the ECB monetary policy decision and the end of the ECB President's press conference). The Target shock is scaled to yield a unit effect on the one-month OIS rate. The QT shock is scaled to yield a unit effect on the ten-year OIS yield.

[1] Unfortunately, the Maturity Ladder does not provide a further breakdown of cash flows that settle beyond five years. Therefore, we had to assign a default maturity value of 15 years to the "Greater than 5 years" bucket, an approach consistent with previous studies (e.g., Coulier et al. (2024)). This assumption is a limitation of our maturity gap measure, as it might cause inaccuracies in the gap distribution across banks. In the worst case, this could misclassify banks with high maturity gaps as having low maturity gaps, and vice versa. However, we are reassured by two key factors. First, the results do not change significantly when we modify the default value (testing for 5, 10, 20, or 30 years). Second, the banks at the tails of our maturity gap distribution (those with the highest or lowest gaps) do indeed have business models that imply a large imbalance between very short-term and long-term cash flows. Hence, we believe our measure still captures valuable information on the cross-sectional differences among banks regarding their exposure to long-term yields.

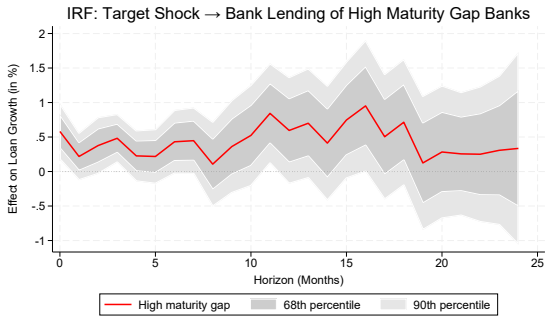
Table 7: Description of control variables

Variable	Unit and Frequency	Data Source	Description
Controls			
Bank size	No unit (log), quarterly	ECB Supervisory Reporting data	Defined as log(Total Assets). Total Assets are the carrying amounts sourced from template F01 of FINREP.
Non-performing loan (NPL) ratio	Percentages, quarterly	ECB Supervisory Reporting data	Defined as non-performing loans over total loans. Only loans granted to the non-financial private sector (i.e., non-financial corporations and households) are considered in both the numerator and the denominator. The carrying amounts are sourced from the FINREP template F18.
Liquidity coverage ratio (LCR)	Percentages, quarterly	ECB Supervisory Reporting data	The liquidity coverage ratio is sourced from the COREP template C76. Banks report this value in line with the definition outlined in Article 4(1) of the Delegated Regulation (EU) 2015/61.
Profitability: Return on assets	Percentages, quarterly	ECB Supervisory Reporting data	Defined as Total Profit/Loss over Total Assets. The numerator is sourced from the FINREP template F02 and adjusted such that it represents the four-quarter trailing sum of profits (i.e., a year-on-year measure). The denominator is sourced from FINREP template F01.
Capital: Common Equity Tier 1 (CET1) ratio	Percentages, quarterly	ECB Supervisory Reporting data	Defined as the CET1 capital over the total risk exposure amount (i.e., risk-weighted assets). This value is reported in COREP template C03, in accordance with point (a) of Article 92(2) of the CRR.
Leverage ratio	Percentages, quarterly	ECB Supervisory Reporting data	Defined by the regulator as the Tier 1 capital amount over the total leverage ratio exposure measure. We use the fully phased-in definition. The denominator includes on-balance sheet assets, securities financing transactions, derivatives exposures, and other off-balance sheet items, net of exemptions (e.g., intragroup exposures, promotional loans,...). The leverage ratio is sourced from the COREP template C47.

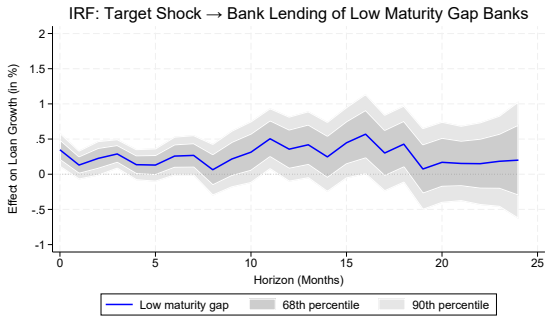
Table 8: Description of additional control variables

Variable	Unit and Frequency	Data Source	Description
Controls			
Loan-to-deposit ratio	Percentages, quarterly	ECB Supervisory Reporting data	Defined as the total loans to the non-financial private sector (i.e., non-financial corporations and households) divided by the total deposits from the non-financial private sector. The carrying amount of loans is sourced from the FINREP template F18, summing the amounts reported under the different accounting rules (i.e., at fair value and amortized cost). The carrying amount of deposits is sourced from the FINREP template F8, which sums the amounts reported under the different accounting rules.
Deposit ratio	Percentages, quarterly	ECB Supervisory Reporting data	Defined as the total deposits from the non-financial private sector (i.e., non-financial corporations and households) divided by total assets. The carrying amount of deposits is sourced from the FINREP template F8, summing the amounts reported under the different accounting rules. Total Assets are carrying amounts sourced from template F01 of FINREP.

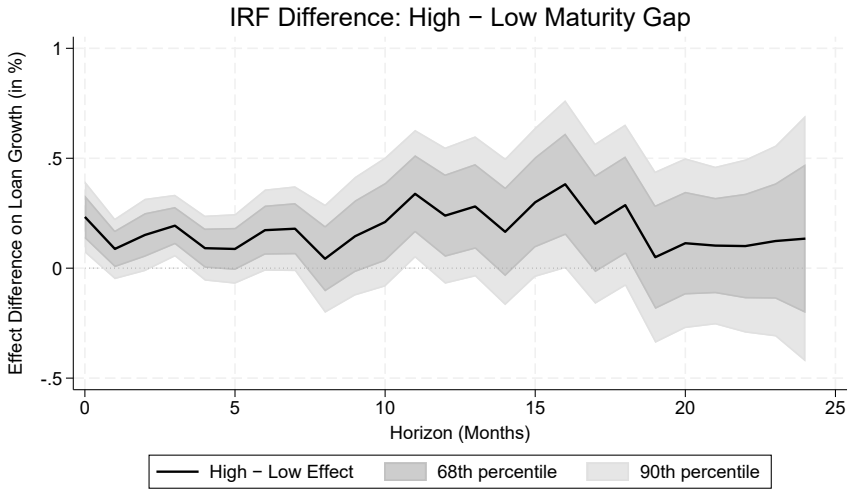
Figure 11: Analysis of bank lending responses to a target shock, comparing high and low maturity gap banks - Full sample



(a) *Response of lending from high maturity gap banks to target shock*

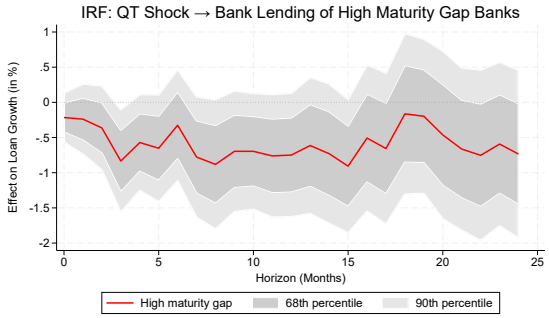


(b) *Response of lending from low maturity gap banks to target shock*

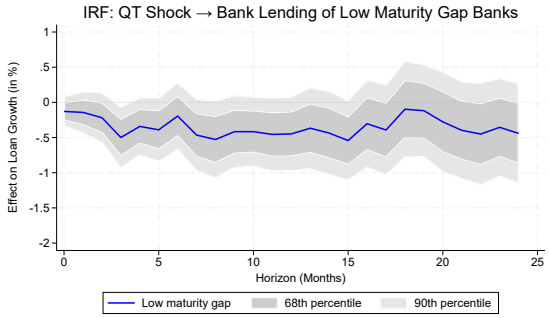


(c) *Difference in bank lending response between banks with high vs low maturity gap under a target shock*

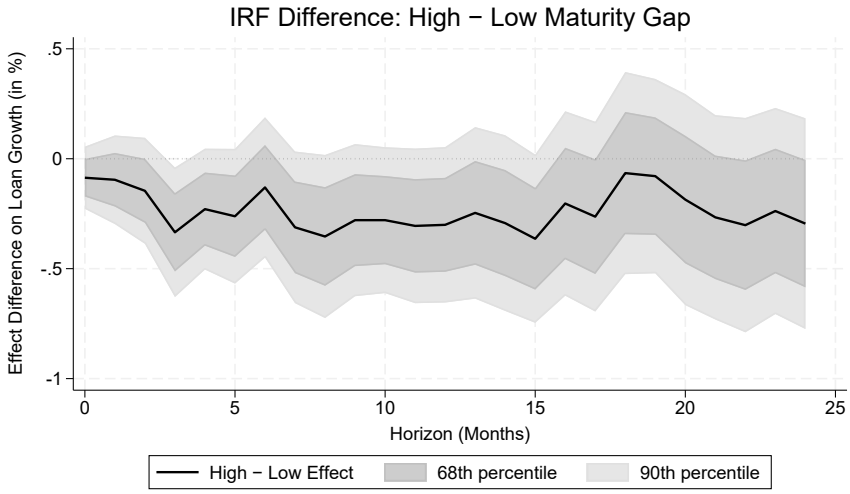
Figure 12: Analysis of bank lending responses to a QT shock, comparing high and low maturity gap banks - Restricted sample



(a) *Response of lending from high maturity gap banks to QT shock*

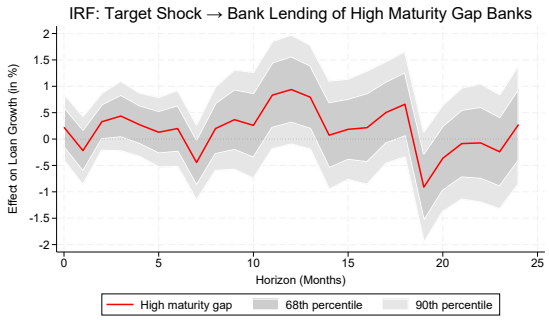


(b) *Response of lending from low maturity gap banks to QT shock*

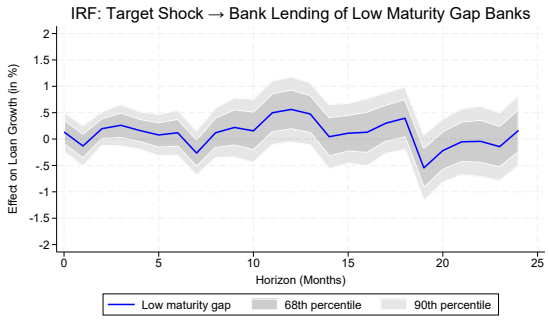


(c) *Difference in bank lending response between banks with high vs low maturity gap under a QT shock*

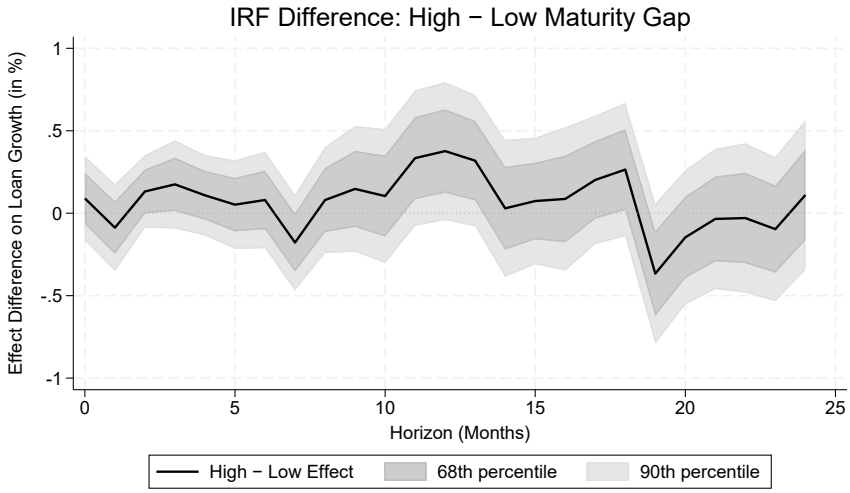
Figure 13: Analysis of bank lending responses to a target shock, comparing high and low maturity gap banks - Restricted sample



(a) *Response of lending from high maturity gap banks to target shock*



(b) *Response of lending from low maturity gap banks to target shock*



(c) *Difference in bank lending response between banks with high vs low maturity gap under a target shock*